

Sustainable Personalised Driving: Applying Human-in-the-Loop Multi-Objective Bayesian Optimisation to Eco-Driving Interface Design

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Abstract

Eco-driving displays are usually evaluated with a few fixed designs identical for all drivers. This study treated display design as a per-participant multi-objective search problem. Sixteen participants each drove 21 short urban rounds in a virtual-reality simulator: 15 Sobol sampling rounds, five proposed by a human-in-the-loop multi-objective Bayesian optimiser over seven continuous display parameters, and one repeat of a recommended configuration. The objectives were a simulated energy-intensity proxy, task load, and three acceptance ratings. In this fixed-order, single-session design, cumulative hypervolume increased for 15 of 16 participants, and the proportion of rounds on a participant's own Pareto front was 0.44 during optimisation versus 0.24 during sampling. The recommended configurations improved all four subjective objectives relative to a same-route sampling baseline, left the energy proxy unchanged, and differed markedly between participants. These fixed-order associations do not establish that personalisation outperforms a strong fixed default.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → *Machine learning*.

Keywords

eco-driving, head-up display, human-in-the-loop optimisation, multi-objective Bayesian optimisation, personalisation, virtual reality

1 Introduction

Road transport is one of the largest and most difficult-to-reduce sources of carbon emissions within the transport sector, and the energy used by the same vehicle on the same road depends substantially on how it is driven [2, 19]. Eco-driving - a set of behaviours including gentle acceleration, anticipatory braking, and maintaining a steady speed - is therefore a relatively low-cost solution: it requires neither a new powertrain nor new infrastructure [2]. In practice, in-vehicle feedback is one strategy for promoting it.

The evidence that feedback is effective is substantial. A meta-analysis of in-vehicle eco-driving feedback combined 23 effect sizes from 17 controlled studies and reported a pooled improvement in fuel economy of 6.6%, with a 95% confidence interval of 4.9% to 8.2% [18].

How feedback is designed matters as much as whether it exists. A moving-base simulator study with 23 lorry drivers recommended

intermittent over continuous presentation because it shortened glances and let designers control where glance frequency increased [15]. Whether a display expresses information numerically or symbolically changes fuel consumption measured in the field [7]. Neither the amount of information nor its modality has an intuitively predictable effect: in a paired comparison of 12 interfaces, a visual display providing more information reduced driver control error, and an added auditory cue improved performance while reducing visual distraction [12]. A design framework for eco-driving feedback categorises information, timing, and presentation as three design dimensions and relates them respectively to the behavioural mechanisms of attention, motivation, and learning [17]. Motivational content based on comparisons with other drivers is intuitively appealing, but did not produce more economical driving when tested [6].

Previous research has generally presented the same display to all participants and compared a small number of predefined conditions. Such designs can estimate average condition effects, but are less able to determine which parameter combination should be used for a particular driver. Earlier work shows that drivers respond differently to goal setting, message framing, and anchoring [20], and personalised eco-driving feedback has also been explored directly [3]. Sumner et al. [21], meanwhile, inferred latent cognitive constructs from recent driving behaviour and used them to trigger a safety interface. Together, these precedents suggest that individual differences should be incorporated into the design process.

Human-in-the-loop Bayesian optimisation offers one candidate method. It treats user ratings and system measurements as expensive, noisy observations of a design space and uses them to select the next design for evaluation [4, 10]. The method has recently been applied to multi-objective automotive-interface optimisation, including visualisations of automated-driving functions, external vehicle communication, and in-vehicle conversational assistants [5, 14, 22]. Eco-driving displays are distinctive because the system must handle both a simulated energy proxy calculated from the driving trajectory and participants' subjective acceptance; the two types of objective differ in their measurement chains, noise, and perceptibility.

This study therefore explores whether an eco-driving display can be treated as a human-in-the-loop multi-objective search problem that proposes candidate configurations online for each driver within a single session. I built an immersive virtual-reality driving simulator reproducing neighbourhoods around Knightsbridge,

Belgravia, and Chelsea in central London, together with a parameterised eco-driving display. Each participant completed 21 rounds on short routes and rated the display after every round; the system used their previous ratings and simulated energy-intensity proxy values to propose subsequent designs.

This study addresses three research questions.

RQ1: In a single session with sampling fixed before optimisation, how do multi-objective quality indicators change between the two phases, and what internal comparison is obtained between the five optimisation rounds and the preceding five sampling rounds?

RQ2: How much do the resulting personalised designs differ between drivers, and which parts of the display show the greatest differentiation?

RQ3: Is a 21-round, single-session optimisation protocol feasible and acceptable to participants in an immersive simulator?

RQ1 is a fixed-order within-session comparison. The study did not include an independent non-adaptive condition, nor were sampling and optimisation rounds randomised or interleaved. Later changes therefore cannot be attributed solely to the optimiser, and the study cannot establish that personalisation is superior to a generic default display. Comparing the five optimisation rounds with the preceding five sampling rounds provides only a budget-matched internal reference and remains confounded by time, practice, rating drift, and different search paths.

Under the contribution types proposed by Wobbrock and Kientz [25], this study primarily makes two contributions. The first is empirical material: a study of per-participant, multi-objective search for eco-driving feedback that reports optimisation indicators across fixed-order phases, participant-associated design differentiation, and experiences of the protocol. The second is an artefact contribution: a parameterised display, self-authored components of a virtual-reality driving simulator, and an implementation connecting them to a multi-objective optimiser. The optimisation algorithm is adopted from previous work; this study does not claim a new optimisation method.

2 Related Work

This section reviews four strands: the effect of in-vehicle eco-driving feedback, the display-design space of information, timing, and presentation, individual differences and personalised approaches, and human-in-the-loop multi-objective optimisation as the basis for the research motivation.

2.1 Eco-Driving and the Effect of In-Vehicle Feedback

Eco-driving is a set of behaviours that reduce the energy a vehicle needs for a given journey: gentle acceleration, anticipatory deceleration, steady speed, and less idling. Barkenbus [2] argues it is undervalued as a climate measure precisely because it needs no fleet replacement or new infrastructure. Sivak and Schoettle [19] decompose drivers' influence into strategic, tactical, and operational decisions; vehicle choice dominates (a ninefold difference between the least and most efficient models on sale), yet the factors under the driver's control can still produce approximately 45% variation in real-world fuel consumption. In-vehicle feedback acts on this final level.

The evidence that feedback works is substantial: Sanguinetti et al. [18] synthesised 23 effect sizes from 17 controlled studies and reported a pooled fuel-economy improvement of 6.6%, 95% CI [4.9%, 8.2%], $p < .0001$, with larger effects for feedback reporting both instantaneous and cumulative performance and diminishing benefits over longer interventions. A review of eco-driving guidance reached a similar conclusion while stressing that use depends on acceptability as well as effectiveness [23]. Effect sizes vary substantially between studies, and part of that variance belongs to the interventions themselves.

This final point motivates the remainder of this section: the effect of eco-driving feedback depends on how the display is designed.

2.2 The Design Space of Eco-Driving Displays

Sanguinetti et al. [17] organise eco-driving feedback design along three axes - what information is displayed, when, and how - identifying them as independent design decisions, each with its own evidence base.

On timing, Kircher et al. [15] compared no feedback, intermittent, continuous, and a driver-configured condition with 23 lorry drivers in a moving-base simulator. Drivers glanced at the system more when traffic was easy and delayed the first glance in complex situations; the authors recommended intermittent presentation because it shortened glances and let designers schedule information where it would not burden the driver. The driver-configured condition is an early form of the problem addressed here.

On presentation, Dahlinger et al. [7] ran a randomised controlled field trial that isolated representational form: symbolic and numerical displays carried the same information and differed only in how it was drawn. Sixty-two roadside-assistance drivers in Switzerland drove for ten weeks and over 245,000 km (265,939 constant-characteristic road segments), randomised to an 'eco-tree' that grew or withered with a three-minute consumption average ($N = 23$), a real-time consumption gauge with pedal positions ($N = 24$), or a basic screen ($N = 25$). Only symbolic feedback significantly reduced fuel consumption, by approximately 0.2–0.3 L/100 km (2%–3%), robust across covariate models; numerical feedback was consistently non-significant, and in the treatment-only model its coefficient was even significantly positive until date was added as a covariate.

On layout, Jamson et al. [12] evaluated 12 eco-driving interfaces by paired comparison in a high-fidelity simulator, delivering pedal-angle advice visually, audio-visually, or haptically, and measuring ratings plus the error between advised and applied pedal angle. Contrary to the intuition that more information is more distracting, a visual display adding the required rate of pedal change reduced pedal error, and an added auditory cue improved performance while reducing visual distraction; among haptic designs, force feedback outperformed stiffness feedback.

On motivational content, Colley et al. [6] added peer comparison to a windscreen eco-driving display in a within-participant virtual-reality study with 19 participants across manual and automated modes. The result was null: peer comparison did not produce significantly more economical driving in either mode, while the driver's own score produced the clearest improvement. Their windscreen layout and eco-score construction are the direct starting points for

the display used here, and their null finding is why the final design has no aggregate peer comparison.

One modality result tempers the assumption that a visual display helps by default. Pietra et al. [16] compared accelerator-pedal haptic feedback with visual feedback on a head-up display (HUD) in a ten-participant virtual-reality simulator: relative to no feedback, haptic feedback reduced the mean power index by 14%, the combination by 11%, and visual feedback alone increased it by 11%. The study was small and not definitive, but it is the closest published setting to the present one and suggests visual eco-driving feedback in virtual reality can worsen energy use.

Taken together, these studies establish three points: eco-driving displays occupy a large design space; different points within that space produce measurably different outcomes; and trade-offs exist between those outcomes. They do not establish which point in the space should be presented to a particular driver.

2.3 Individual Differences and Personalised Feedback

The studies above compare a few predefined designs and report average differences, answering which display is best on average - not whether that display is best for a particular driver, and there are good reasons to expect it often is not.

Stillwater and Kurani [20] interviewed 46 plug-in hybrid drivers across feedback-off and feedback-on phases of a multi-week field trial; three quarters changed how they drove, at least short-term. Two findings matter here: 'personalization of information allows different drivers to access information meaningful to them, thus increasing its motivational value', and instantaneous feedback was used mainly to experiment with new behaviours whereas average feedback was used to set and achieve goals. The study relied on self-report. Brouwer et al. [3] directly explored personalised eco-driving feedback and provide the closest precedent.

Their within-participant study put 26 professional lorry drivers through a baseline, a basic display, and displays customised for learning-oriented or performance-oriented drivers in the TNO truck simulator. Group means were directionally consistent with personalisation: learning-oriented drivers rated both customised displays more useful, and performance-oriented drivers rated the learning-oriented display least acceptable. The authors nevertheless called the acceptance gain an 'increase that could not be validated', judged the sample underpowered, and estimated an adequate between-group test would need at least 80 participants. Directly relevant here, they excluded simulated fuel consumption because 'it is difficult to obtain realistic fuel consumption data in a driving simulator'. The study supports the premise that personalisation may be needed, not that it works for energy use.

More recent work personalises by first inferring driver characteristics: Sumner et al. [21] trained a recurrent neural network to infer impulsivity and inhibitory control from recent driving and used them to trigger a safety interface. This is powerful when a reliable inference model exists, but the system can then personalise only along dimensions its driver model represents, and building that model needs independent data collection and validation.

An alternative is to omit the driver model entirely and optimise the interface directly from the driver's own responses. This is what human-in-the-loop optimisation does.

2.4 Human-in-the-Loop Optimisation of User Interfaces

Human-in-the-loop optimisation treats the design space as a search space and users' responses as observations of an unknown objective function. Bayesian optimisation suits this setting because it was designed for expensive, noisy evaluations - precisely what asking a person to use and rate an interface is [10]. A surrogate model, usually a Gaussian process, is fitted to the observations, and an acquisition function selects the next design by balancing exploration against exploitation.

Chan et al. [4] had 40 novice designers optimise a three-dimensional touch-interaction technique: the optimiser improved design-space coverage, solutions, and mental effort, but guided designers reported lower agency, expressiveness, and creativity, so the authors judged the method suitable where agency is not critical. The trade-off is directly relevant here because my participants rated designs selected for them and had no control over the sequence.

When several outcomes matter and cannot be combined without arbitrary weights, the problem becomes multi-objective, and optimisation returns a Pareto set in which no design can improve one objective without worsening another. Daulton et al. [8] made this tractable with a differentiable expected hypervolume improvement acquisition function, implemented in BoTorch [1]; hypervolume, the objective-space volume dominated by the Pareto set relative to a reference point, gives a scalar summary of the trade-off surface for tracking progress.

Applied to automotive interfaces, Jansen et al. [14] optimised automated-driving visualisations against six user-experience objectives in a 117-participant between-participant online study, comparing one cold-start and two warm-start human-in-the-loop conditions with expert designs, user-built designs, and a control. Optimisation significantly outperformed expert and user designs in perceived safety, trust, and predictability with lower cognitive load, but not in acceptance or aesthetics - a null worth noting since three of the five objectives here measure acceptance. Colley et al. [5] applied the method to external communication of automated vehicles, and Susak et al. [22] extended it to an in-vehicle conversational assistant.

OptiCarVis also observed that under optimisation many parameters were pushed towards off, in contrast to expert and manual configurations [14]. An optimiser given the option to remove elements will use it, and the present parameterisation, in which an element disappears as its size approaches zero, provides exactly that option.

Two points stand out for eco-driving: all objectives in these studies were subjective, whereas the environmentally consequential objective here is energy, measured by the vehicle and affected only indirectly through behaviour; and none had to reconcile an objective behavioural measure with subjective acceptance inside one optimisation loop.

2.5 Summary and Research Motivation

The literature supports four relatively robust observations. First, in-vehicle eco-driving feedback can improve fuel economy on average [18, 23]. Second, its effect depends on design decisions concerning information, timing, and presentation [7, 12, 15, 17]. Third, drivers differ in how they interpret and respond to the same feedback, and personalisation has been explored directly [3, 20, 21]. Fourth, human-in-the-loop multi-objective Bayesian optimisation has been used to search automotive-interface design spaces under limited user-evaluation budgets [4, 5, 14, 22].

Together, these studies raise a question that still requires methodological exploration: how to adapt an eco-driving display online with the driver in the loop while considering both an energy objective calculated from the driving trajectory and subjective acceptance. Prior research has separately addressed personalised eco-driving feedback, human-in-the-loop automotive-interface optimisation, and objective driving outcomes; this study explores their combination as an implementation to be tested.

Accordingly, this study examines changes in multi-objective indicators within a fixed evaluation budget (RQ1), differentiation between the resulting participant configurations and its possible sources (RQ2), and the feasibility and experience of a 21-round virtual-reality protocol (RQ3). The research questions and conclusions are limited by the fixed phase order, absence of a strong fixed baseline, and use of a simulated energy proxy.

3 Method

3.1 Study Design

The study used a within-participant design: each participant experienced a sequence of display designs from the same design space, determined by a per-participant optimisation loop rather than a fixed condition order, over 21 driving rounds. The first 15 rounds formed a quasi-random space-filling phase, the next five were proposed by a multi-objective Bayesian optimiser fitted to that participant’s own observations, and the final round evaluated the design recommended by the optimiser at the end of the session.

The independent variable was the display configuration, a point in a seven-dimensional continuous design space. Five dependent variables were returned as objectives: energy use for the round and one task-load score (minimised), and three independent acceptance ratings (maximised).

The study did not include an independent non-adaptive baseline, and the sampling phase was fixed before the optimisation phase. The results therefore describe only associations across fixed-order phases and budget-matched internal comparisons within the same design space and participant, not an independent causal effect of the optimiser relative to time, practice, rating drift, or search paths.

3.2 Participants

Sixteen participants took part (P1–P16). They were 22–25 years old ($M = 23.1$, $SD = 0.9$) and had held a driving licence for 2–7 years ($M = 3.6$, $SD = 1.5$); ten were men, five were women, and one was non-binary. Four drove almost daily, three weekly, four monthly, and five rarely or never. Four had never used a VR



Figure 1: Experimental apparatus: Yaw3 three-degree-of-freedom motion platform, racing seat, Logitech force-feedback steering wheel, and three-pedal set. Participants sat wearing a Meta Quest 3S head-mounted display (the unoccupied apparatus is shown).

headset, nine had tried one several times, and three were regular users; eleven had no experience of a steering-wheel simulator. Demographic information was collected in the post-study questionnaire. No completed session was excluded. No a priori power analysis was conducted; the sample size was constrained by the available recruitment window and the duration of the 21-round protocol for each participant. Participants were recruited through an open advertisement on a supervisor-maintained Discord server (members primarily students) and direct approaches by the author to acquaintances. Participation was unpaid, with no payment or voucher offered, as stated in the information sheet before consent.

3.3 Experimental Apparatus

3.3.1 Simulator Hardware. Participants wore a Meta Quest 3S head-mounted display connected to the host computer by a wired link, with rendering performed on the host. Steering and pedal input were provided by a Logitech G25 steering wheel and pedal set. Inertial cues were presented by a Yaw VR motion platform, to which the simulator sent pose updates directly through the device’s network protocol (Figure 1 and Figure 2).

After the eighth participant, the pitch axis of the motion platform failed because the rubber on the friction wheel split and the pitch motor malfunctioned; it could not be repaired within the study period. P1–P8 therefore experienced inertial cues, whereas P9–P16 drove with the platform stationary. Table 1 gives the complete host, headset, and runtime configuration. Frame rate came from the Meta runtime’s own telemetry logs; during periods overlapping the sessions, the mean was 71.99 fps, close to the 72 Hz display limit.

3.3.2 Virtual Environment. The simulator was built in Unity 6000.3.9f1 and reproduces Knightsbridge, Belgravia, Chelsea, and surrounding streets in central London. The visual streetscape was streamed at runtime from Google Photorealistic 3D Tiles

Table 1: Simulator computer, software, virtual-reality and motion-platform configuration, and measured frame rate during the experiment. Hardware and runtime values were taken from Meta runtime performance logs, the memory-module layout was obtained from a system query, and the engine version was taken from the project settings.

Category	Item	Configuration
Host computer	Processor	AMD Ryzen 5 9600X (6 cores)
	Memory	2 x 32 GB DDR5-5600 (63,063 MB available)
	Motherboard	ASRock B650M Pro RS WiFi
	Graphics card	NVIDIA GeForce RTX 4070 Ti SUPER, 16 GB, driver 571.96
Software	Operating system	Windows 11 64-bit (build 26200.9168)
	Engine	Unity 6000.3.9f1
	VR runtime	Meta runtime 1.207.0 (OpenXR)
Virtual reality	Head-mounted display	Meta Quest 3S, direct display connection
	Refresh rate	72 Hz
	Connection	Wired Quest Link (USB3), encoding resolution 3680 x 1920
	Per-eye rendering resolution	1832x1920
Motion platform	Model and interface	Yaw3, 50 Hz pose stream, motion condition only
Measured frame rate	Within-session mean	71.99 fps
	Worst 60-second bin mean	70.42 fps
	5th-percentile bin mean	71.37 fps
	Compositor dropped frames	0.0587%
	ASW-synthesised frames	0%
	Telemetry coverage	35 telemetry bins, 73,245 frames, 16.96 minutes



Figure 2: Side view of a participant during the experiment, with the face pixelated. The head-mounted display is connected by a wired Link cable, both hands are on the steering wheel, both feet are on the pedals, and the motion platform is in its operating position.

through Cesium for Unity 1.23.3, while the surface participants actually drove on was a separately constructed road mesh based on OpenStreetMap geometry. The driveable network comprised 418 road segments with a total centreline length of 28.7 km and 564 junctions: 486 three-way junctions, 76 four-way junctions, and two five-way junctions.

Ambient traffic was simulated across the network with a commercial traffic system (Gley Traffic System v3.6.2), using priority control at junctions and traffic-light control at major junctions. Static pedestrians were placed on pavements. Participants drove a passenger car from a seated, first-person cockpit view. The interior

rear-view mirror and both exterior mirrors rendered live rearward views onto the vehicle model’s own mirror surfaces.

3.3.3 Vehicle Dynamics. Vehicle control was configured for steering-wheel and pedal input. The vehicle used a torque curve, a maximum speed of 20 m/s (72 km/h), a driving force of 6500, a braking force of 9000 (simulator units), and a drag coefficient of 0.1. In practice, its terminal speed on level ground was approximately 52 km/h.

3.4 Eco-Driving Display and Design Space

The display comprised two types of spatial object. A leaf gauge, numerical eco-score, valence icon and advice text (jointly termed the framing element), speedometer, and acceleration icon were placed on a world-space canvas at the windscreen; vehicle markers were attached above surrounding traffic vehicles and faced the driver. A route-guidance arrow showing the next turn and its distance was always present and was not part of the design space.

The design space comprised seven continuous dimensions. Six independently set the size of one display element, and the seventh set overall display opacity. The optimiser operated on a normalised unit interval for each dimension, while the mapping from that interval to physical values was encapsulated in the display component.

Table 2 gives the physical mappings. An element was not drawn when its physical scale fell below 0.6, a normalised threshold of $0.6/1.3 \approx 0.462$; the speedometer was always retained and kept its own minimum readable alpha.

Two consequences follow. First, element presence is controlled by the same continuous variable as size, which keeps the search space fully continuous, avoids mixed-variable Bayesian optimisation under a small budget, and still lets the optimiser remove any element except the speedometer by shrinking it to zero. Second, opacity controls overall salience while colour coding stays at full intensity, so a dim display is uniformly dimmed as a whole.

Table 2: Seven-dimensional design space. Each dimension was normalised to the unit interval for the optimiser, while the physical mapping was encapsulated in the display component.

Dimension	Element controlled	Optimiser range	Physical mapping	Can vanish
size 1	Leaf gauge	0–1	x 1.3 (scale)	yes
size 2	Numeric score	0–1	x 1.3 (scale)	yes
size 3	Framing face + advice line	0–1	x 1.3 (scale)	yes
size 4	Speedometer	0–1	0.6 + 0.7x (scale)	no (research-design floor)
size 5	Acceleration pictogram	0–1	x 1.3 (scale)	yes
size 6	Vehicle markers	0–1	x 1.3 (scale)	yes
opacity	Whole display alpha	0–1	0.10 + 0.90x	no (floor 0.10)

Colour coding within the display represents the driver’s current eco-state: as the components of the eco-score change, the speedometer and acceleration icon transition between green, amber, and red.

3.5 Eco-Score and Energy Objective

The study used two distinct quantities that must not be conflated. The eco-score was visible to the driver during a round. The energy objective was returned to the optimiser after the round ended.

The eco-score construction followed Colley et al. [6]. The acceleration sub-score used the piecewise-linear form described in their Equation 2: the score was 100 when $|a| < 0.01 \text{ m/s}^2$, zero above a cut-off, and decreased linearly between the two. The printed equation gives a slope of 33.9 and a cut-off of 2.95 m/s^2 , which are not fully consistent with the adjacent statement that ‘a speed change of 100 km/h completed in 10 s should score zero’. The latter converts to 2.78 m/s^2 and corresponds to a slope of approximately 36.1. The present implementation followed that textual assumption and used a cut-off of 2.78 m/s^2 and a slope of 36.1.

In this study, the speed sub-score was reconstructed as an asymmetric Gaussian centred on the target speed, given in Equation 1.

$$V(v) = 100 \exp\left(-\frac{(v - v^*)^2}{2\sigma^2}\right), \quad \sigma = \begin{cases} \sigma_{\text{low}} & v \leq v^* \\ \sigma_{\text{high}} & v > v^* \end{cases} \quad (1)$$

where $v^* = 45 \text{ km/h}$, $\sigma_{\text{low}} = 14 \text{ km/h}$ on the low-speed side, and $\sigma_{\text{high}} = 4.5 \text{ km/h}$ on the high-speed side.

The curve is shown in Figure 3. Its asymmetry was intended to impose only a gradual penalty below the target speed and a steeper penalty above it. Table 3 gives representative values.

The energy objective supplied to the optimiser was independent of the eco-score visible to the driver. It was an energy-intensity proxy based on simulated road load. At each time step, the model computed $F = ma + C_r mg + \frac{1}{2} \rho C_d A v^2$ and $P = Fv$. When $P \geq 0$, the energy increment was $P \Delta t / \eta_d$; when $P < 0$, it was $\eta_r P \Delta t$. The implementation used $m = 1500 \text{ kg}$, $C_d = 0.30$, $A = 2.2 \text{ m}^2$, $C_r = 0.012$, $\rho = 1.225 \text{ kg/m}^3$, $\eta_d = 0.90$, $\eta_r = 0.60$, and $g = 9.81 \text{ m/s}^2$, and the result was converted to kWh/100 km using the distance driven. The model included no explicit potential-energy term for road gradient and no separate collision-energy term; collisions entered only indirectly through the speed and acceleration trajectory. The parameters were not calibrated against a real vehicle. The measure is therefore interpreted only as an internally consistent simulated proxy. The optimisation bounds were 0–150 kWh/100 km; the upper bound was widened after truncation occurred in an early pilot.

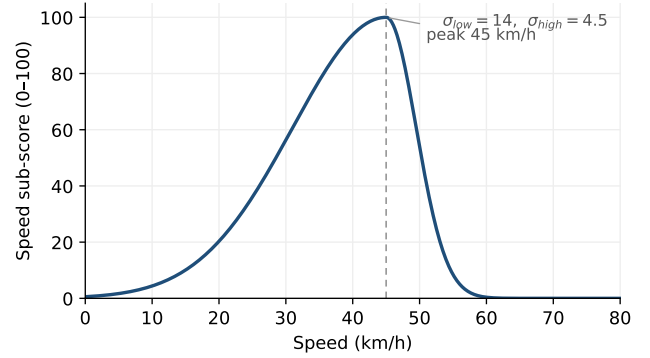


Figure 3: Speed sub-score curve: an asymmetric Gaussian peaking at 45 km/h, with $\sigma = 14 \text{ km/h}$ on the low-speed side and $\sigma = 4.5 \text{ km/h}$ on the high-speed side.

3.6 Optimisation Setup

Optimisation used the differentiable expected hypervolume improvement (qEHVI) acquisition function introduced by Daulton et al. [8] and implemented in BoTorch [1]. Bayesian Optimization for Unity [13], an open-source toolkit, bridged simulator and optimiser by running BoTorch as a separate process on the host and exchanging parameters and objectives over a local socket.

The optimisation problem had seven design dimensions and five objectives (Table 4). The first 15 rounds were Sobol samples from the unit hypercube, followed by five qEHVI rounds with a batch size of one. Round 21 repeated a candidate already observed during the first 20 rounds and did not propose a new design. The selector first chose from the observed Pareto set by minimising normalised distance to the ideal point; ties were resolved in sequence by maximising the minimum margin, minimising aggression, and choosing the earlier round. Of the 16 recommended configurations, eight originated in the sampling phase and eight in the optimisation phase. The acquisition function used 512 Monte Carlo samples, 1,024 raw samples, and ten restarts. The 15 initial points followed the ‘twice the dimensionality plus one’ heuristic. Compared with the five sampling and ten optimisation rounds used by OptiCarVis, this study allocated more of its budget to covering the seven-dimensional space and consequently retained only five model-driven iterations.

Because elements hid as their sizes approached zero, Sobol sampling could in principle produce a configuration with every optimiser-controlled element invisible; such samples were detected

Table 3: Speed sub-scores at representative speeds.

Speed (km/h)	25	30	32.2	40	45	47	50	53
Sub-score	36.0	56.3	65.8	93.8	100.0	90.6	53.9	20.6

Table 4: The five optimisation objectives, their bounds, directions, and sources.

Objective	Bounds	Direction	Source
Simulated energy proxy (kWh/100 km)	0–150	minimise	road-load integral over the driven round
Task load	0–100	minimise	mean of survey items 1–2
Informativeness	0–100	maximise	survey item 3
Pleasantness	0–100	maximise	survey item 4
Legibility	0–100	maximise	survey item 5

and repaired before presentation, so no participant experienced a round with no eco-driving feedback at all.

3.7 Measures

At the end of each round, participants answered five items about that round’s display using 21-position discrete sliders labelled from 0 to 20 and marked with tick marks (Table 5).

The three acceptance items were kept as separate objectives because the same design change could move informativeness, pleasantness, and legibility in opposite directions, and averaging would conceal the conflict the optimiser was meant to negotiate. Load items were worded so that higher values were undesirable and acceptance items so that higher values were desirable, avoiding a built-in positive correlation. Usefulness was operationalised as informativeness because fuel consumption could not be perceived directly in the simulator.

One data-quality constraint applied: each slider opened at the midpoint and the submit control stayed disabled until every slider had been moved at least once, distinguishing an intentional midpoint rating from a missing response; slider positions did not carry over between rounds. Objective driving data, including speed, acceleration, components of the eco-score, and energy use, were recorded throughout every round.

After the experiment, participants completed a post-study questionnaire in Qualtrics, UCL’s survey platform. It collected demographics and prior experience: age, gender, years since obtaining a driving licence, driving frequency, and experience with VR headsets and steering-wheel simulators. It further contained five agreement items on five-point scales about the final display design, one five-point item on dizziness, nausea, or eye strain, and one item on the experience of seat movement. Two open questions asked which display element they liked and disliked most and whether the final design was their ideal design.

3.8 Driving Task and Routes

Each round consisted of one drive along one of four specified urban routes; their positions and starting points in the road network are shown in Figure 4. R1 was 215 m long and R2 was 165 m long, and both were straight. From R2 onwards, stationary pedestrians

were added to the pavements. R3 was 283 m long and included one turn greater than 30°; R4 was 170 m long and included three turns. A windscreen guidance arrow indicated the next turn and the remaining distance. Participants drove among ambient traffic and traffic lights, and the round ended when the route was completed. A round was terminated early if its duration exceeded a route-length-scaled limit, preventing a stuck vehicle from halting the entire session.

3.9 Experimental Procedure

After an introduction and written informed consent, participants put on the head-mounted display, familiarised themselves with the vehicle controls and motion platform, and completed the 21 rounds, with the five-item questionnaire presented in the virtual environment immediately after each round. The experiment concluded with the post-study questionnaire and a brief debriefing.

The median duration of the in-simulator portion (driving and per-round questionnaires) was 24.2 minutes (IQR 18.2–32.1, range 16.5–38.0). Under a separate component-based accounting of the logs, mean driving time was approximately 14.9 minutes, mean questionnaire time approximately 10.4 minutes, and cumulative mean optimisation-computation time approximately 6.7 minutes.

3.10 Pilot Calibration

The author completed one 36-round manual pilot to verify the workflow and set approximate energy bounds. The minimum pilot observation was 44.8 kWh/100 km; four of the 36 rounds were truncated at the original upper bound of 100 before being written to disk, so their true values cannot be recovered, and the bound was widened to 150 for the formal study. This was an engineering adjustment made after seeing the pilot results, not a preregistered validation, and the pilot preceded finalisation of the vehicle dynamics and motion platform; its role was limited to identifying truncation risk and confirming that the workflow could run.

3.11 Ethics and Data Protection

The study was approved by the UCL Research Ethics Committee under project number 1165. Before the experiment, participants

Table 5: Five items in the per-round questionnaire. The panel title was “Rate THIS round’s display”, with scale anchors from ‘0 = not at all’ to ‘20 = very much’. Each item used a 21-position integer slider that opened at the midpoint of 10. Items 1 and 2 were averaged to form the task-load objective, while Items 3–5 each formed a separate acceptance objective. The four subjective objectives were linearly mapped to 0–100 before being passed to the optimiser. Single-item measures were used to keep the per-round questionnaire within a duration sustainable for the optimisation loop.

#	Item (wording shown to participants in VR)	Range	Construct	Optimisation objective	Direction	Source
1	It was mentally demanding	0–20	Mental demand	Task load	Minimise	NASA-TLX mental-demand subscale
2	It distracted me from the road	0–20	Attentional interference	Task load	Minimise	Author-developed; detects interference with the primary task
3	It kept me informed about my driving	0–20	Usefulness, operationalised as informativeness	Informativeness	Maximise	Based on the van der Laan acceptance scale
4	It was pleasant to use	0–20	Satisfaction	Pleasantness	Maximise	Based on the van der Laan acceptance scale
5	I could read it at a glance	0–20	Glance legibility	Legibility	Maximise	Author-developed for glance-based use of a head-up display

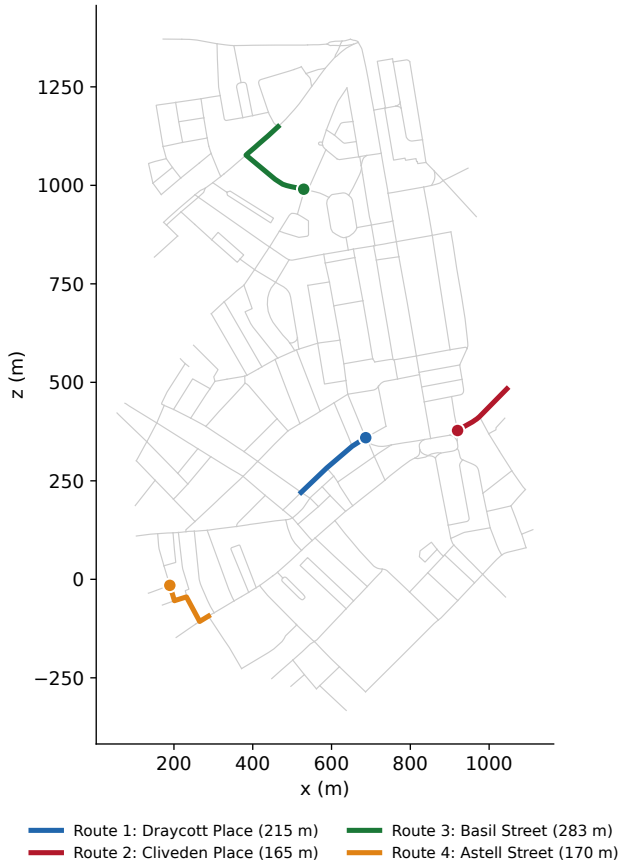


Figure 4: Driveable road network (grey), the four experimental routes, and their starting points.

received a participant information sheet and signed a written consent form; the task instructions and the content of both documents are reproduced in Appendix G. The study was conducted in the

laboratory. The information sheet explained that virtual reality could cause nausea, eye strain, and disorientation and specified responses including stopping the study, removing the headset, and resting if discomfort occurred. Participation was voluntary and unpaid; participants could withdraw without giving a reason and could specify how data already collected should be handled. The participant shown in Figure 2 separately consented to research photography and use of the photograph in the thesis; the face has been pixelated in this draft.

Exclusion criteria covered a history of binocular-vision abnormalities, mental illness, heart disease, implanted medical devices, seizures, photosensitivity, or severe vertigo. These criteria were stated in the information sheet and confirmed on the consent form.

Research data were stored under pseudonymous identifiers, and the thesis uses labels such as P1 and P2. The existing materials are not fully consistent in their descriptions of open access: the participant information sheet states that anonymised data may be made public, whereas the consent form states that pseudonymised data may be used by authenticated researchers in future research. Per-round data and high-dimensional trajectories are therefore not described as already public before the principal investigator and ethics review confirm the permitted scope; any release must first undergo a re-identification-risk assessment and comply with the approved data-governance arrangements.

3.12 Analysis

The unit of analysis was the participant, with rounds as repeated measures: every inferential comparison used one value per participant or per participant per phase, and the 336 mutually dependent rounds were never treated as independent observations. Phase had three levels: sampling (Rounds 1–15), optimisation (Rounds 16–20), and final design (Round 21), with the final design being a repeat drive of a design evaluated in an earlier round. Analysis was conducted in R 4.6.1. ARTool provided the aligned rank transform analysis of variance used for omnibus phase tests. moocore and emoa independently computed Pareto fronts and hypervolume and were used to cross-check one another. lme4 and lmerTest fitted

linear mixed models and supplied significance tests for random-slope likelihood-ratio comparisons. `emmeans` computed estimated marginal means; `effectsize` computed effect sizes and their intervals; `rmcorr` computed within-participant repeated-measures correlations; and `colleyRstats` formatted the results consistently. Package versions were `ARTool` 0.11.2, `moocore` 0.3.2, `emoa` 0.5-3, `lme4` 2.0-6, `lmerTest` 3.2-1, `emmeans` 2.0.4, `effectsize` 1.0.3, `rmcorr` 0.7.0, and `colleyRstats` 0.1.6. Continuous variables were summarised with means and standard deviations. Non-parametric or rank-based tests were used throughout because the participant-level sample was small (16, or eight per group after grouping), a size at which normality could not be established. Exact *p* values are reported to three decimal places.

RQ1 was examined from four perspectives, since no single summary suffices for a five-objective problem. First, an exact two-sided Wilcoxon signed-rank test compared hypervolume after the final optimisation round with that at the end of sampling, reporting *V*, the paired rank-biserial correlation, and paired-samples Cohen's *d*. Second, because cumulative hypervolume can rise simply from drawing five more designs, the gain over the five optimisation rounds was compared within participant with the gain over the preceding five sampling rounds. Cumulative hypervolume was re-computed after each evaluation in the normalised minimisation space, with reference point (1, 1, 1, 1, 1). Third, each objective was aggregated to one value per participant per phase; an aligned rank transform (ART) analysis of variance tested the omnibus phase effect, with a Friedman test as rank-based cross-check. No pairwise post-hoc comparisons were calculated, so both are reported only as evidence that phase mattered. Fourth, per-round Pareto hit rates were compared between the two phases. A round was Pareto-optimal when no other round for the same participant was no worse on all five objectives and strictly better on at least one. Fronts were calculated independently with `moocore` and `emoa`, and both implementations were checked against the optimiser's recorded flag for every round.

The final recommended design was compared with a route-matched sampling baseline, the participant's sampling round on the Round 21 route; matching controlled route identity only, not phase order, practice, traffic state, rating drift, or within-route driving differences. The five comparisons were Holm-corrected across the five objectives within the same family, with both raw and adjusted *p* values reported.

RQ2 was examined from three perspectives. First, between-participant dispersion was reported for each dimension and compared with the mean within-participant standard deviation over the five optimisation rounds. Second, a one-sided label-permutation test (5,000 permutations) asked whether pooled Pareto designs were closer to others with the same participant label than under random relabelling; it cannot separate preference from initial Sobol points and search paths, so the result is described as participant-associated clustering. Random-slope likelihood-ratio tests then examined whether parameter-objective associations varied by participant. The transfer analysis used ridge-regression response surfaces, but was interpreted exploratorily only after leave-one-round-out predictive validity had been reported.

RQ3 was summarised descriptively using experimental logs and the post-study questionnaire. The motion and no-motion groups

were compared using exact two-sided Mann-Whitney tests on session-level summaries and were not pooled before this check. All analyses were run by scripts from the raw logs through to every reported value.

The sample size was not determined by an a priori power analysis, but by the available recruitment window and the duration of the 21-round protocol for each participant; a sensitivity analysis is therefore reported. With two-sided $\alpha = .05$ and power of .80, the minimum detectable effect for a within-participant paired comparison ($N = 16$) was $dz = 0.75$, or approximately 0.77 when using the signed-rank test. For comparisons between the motion and no-motion groups (eight participants per group), the minimum detectable effect was $d = 1.51$, or approximately 1.54 with a Mann-Whitney test. The design was therefore sensitive to at least moderate within-participant effects, but only to very large between-group effects.

4 System Implementation

This section describes the simulator and optimisation pipeline because engineering decisions directly delimit what could be measured; the Open Science section states what the public repository does and does not contain.

4.1 Environment Construction: Three-Dimensional City Tiles and Driveable Roads

The environment combined two geometry sources. Google Photorealistic 3D Tiles, streamed through Cesium for Unity 1.23.3, provided the neighbourhood's visual identity; being photogrammetric reconstructions - uneven at wheel scale, with vehicles and vegetation baked in and no clean collision geometry - they served visual presentation only. The driveable surface was generated separately by CityGen3D from OpenStreetMap data, providing geometrically correct carriageways, kerbs, and pavements, and vehicle physics operated entirely on it.

The tile set was georeferenced and rigidly aligned with the road network, and the photogrammetric surface was clipped along a corridor above the carriageway with a height threshold so that overhanging canopies and buildings remained (Figure 5, Figure 6, and Figure 7).

4.2 Windscreen Display

The HUD comprised a world-space canvas at the windscreen - carrying the leaf gauge, numerical eco-score, framing element, speed group, and acceleration icon - and world-space markers above surrounding vehicles, shown or hidden by viewing aperture, occlusion, and distance. The seven design parameters controlled the six element scales and overall opacity (Figure 8 and Figure 9); three in-vehicle mirrors showed live rearward views through render textures.

4.3 Motion-Platform Integration

The motion platform applied tilt coordination to low-pass-filtered longitudinal and lateral acceleration, with pitch limited to $\pm 12^\circ$, roll to $\pm 10^\circ$, rate to $20^\circ/\text{s}$, washout at $4^\circ/\text{s}$, and yaw following vehicle



Figure 5: Overview of the virtual environment, looking directly down over the Knightsbridge area. Google Photorealistic 3D Tiles streamed through Cesium provide the neighbourhood appearance.



Figure 6: The same viewpoint with the tiles disabled, leaving only the driveable road surface, kerbs, and pavements generated by CityGen3D from OpenStreetMap data. Vehicle physics operated entirely on this surface.



Figure 7: Forward view while driving.

heading. Target angles streamed to the controller over UDP at 50 Hz; TCP served only connection establishment and writing device-side

travel limits, which were enforced by the device itself (Figure 10). The link was tested on the manufacturer’s desktop simulator before validation on the physical platform.

4.4 Vehicle Physics on a Height-Map Road Surface

The driveable surface was a terrain height map, so junctions and crowns had physical gradients. The body collider was narrowed and raised, handling only collisions with walls and vehicles, while four low-friction spherical colliders at the wheel positions supported the vehicle over gradients, dropped kerbs, and tile seams.

Energy use was integrated along the trajectory using the road-load model described in subsection 3.5 and converted to kWh/100 km. It was independent of the eco-score visible to the driver and was used only as a simulated energy-intensity proxy.

4.5 Vehicle Control and Steering-Wheel Input

The wheel and pedals were read through the operating system’s native joystick interface rather than the engine’s input layer; an initialisation component set native mode with 900° rotation and independent axes. A dead zone in the raw range prevented slight off-centring from producing steering input, the paddles acted as a forward/reverse two-speed gearbox, and analogue and keyboard inputs were combined by greater magnitude so the experiment could continue if the wheel failed.

4.6 Ambient Traffic and Its Behaviour

Ambient traffic ran on a commercial system over a network built programmatically from the exported road graph. Each segment generated an open road; junctions received priority control with stopping waypoints and position-matched lane connections; and major junctions along the routes received signals and stop lines. Three additional behaviours were required for these narrow, two-way streets.

First, speed variation: each vehicle carried an individual cruising factor sampled below the road’s configured speed and resampled at staggered intervals, so speeds differed and drifted without exceeding the road setting.

Second, overtaking was disabled: the native behaviour steered into the opposing lane, effectively against traffic. It was replaced by queue dispersal - vehicles stuck off-camera behind the player and moving slowly beyond a threshold were silently removed and respawned elsewhere after an overlap check.

Third, head-on deadlocks were resolved: all vehicle pairs were scanned every 0.3 s, and a pair counted as an imminent encounter when its heading dot product was below -0.6 , separation below 28 m, distance still closing, and extrapolated lateral separation below 2.6 m; both were then pulled towards their lane edges and slowed to pass.

Finally, a 30 m bubble was cleared around the player at each round start, the queue disperser removed followers, and the encounter resolver cleared deadlocked pairs. An encounter-guarantee mechanism therefore ensured that at least one ambient vehicle appeared in view during every round, keeping the marker dimension meaningful (Figure 11).

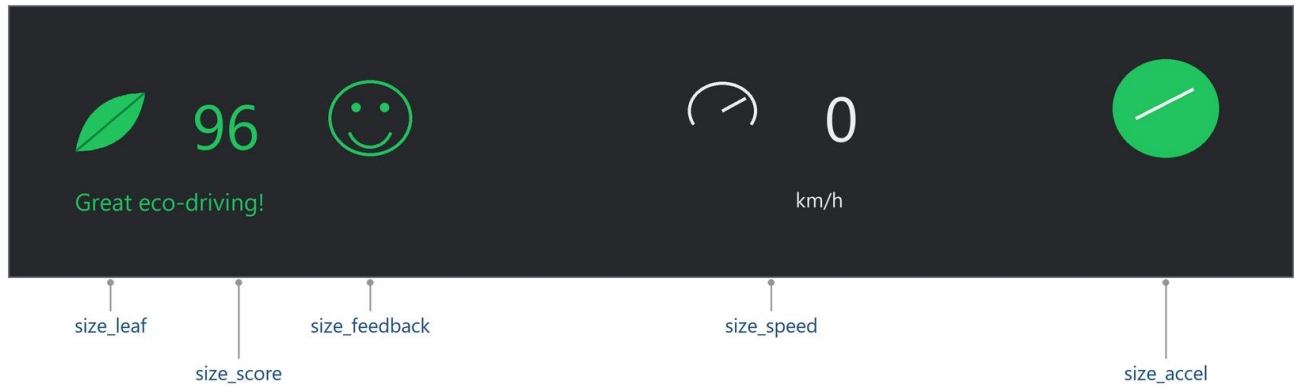


Figure 8: Six adjustable element types in the eco-driving display. The illustrated windscreen canvas shows five static channels: the leaf gauge, numerical eco-score, framing element (valence face and advice text), speed group, and acceleration icon. The sixth type comprises world-space markers drawn above surrounding vehicles at runtime and is not part of this static canvas.



Figure 9: Effects of the size and opacity mechanisms, captured at the same location, under the same lighting, with the vehicle stationary. Top left: elements at a large size. Top right: the same elements at a smaller size. Bottom left: optimiser-controlled elements below the hiding threshold, leaving only the speedometer and the route-guidance arrow on the instrument cluster, neither of which can be removed by the optimiser. Bottom right: the entire canvas at its minimum opacity, with all elements uniformly dimmed while colour coding remains at full intensity; the speed reading remains legible because it has its own readability floor.

4.7 Experimental Loop

A round controller drove the experiment: it applied the candidate design, placed the vehicle at the route start, ran the drive, presented the in-VR questionnaire, returned the five objective values, and received the next candidate over the local socket. Each round wrote the seven parameters, five objectives, phase, and Pareto membership to a per-participant log. A development-only automated-driving

surrogate replaced the participant during testing and generated no research data.

4.8 Implementation of the Questionnaire in Virtual Reality

After each round, the questionnaire appeared as a world-space canvas above the bonnet, with the windscreen display still visible

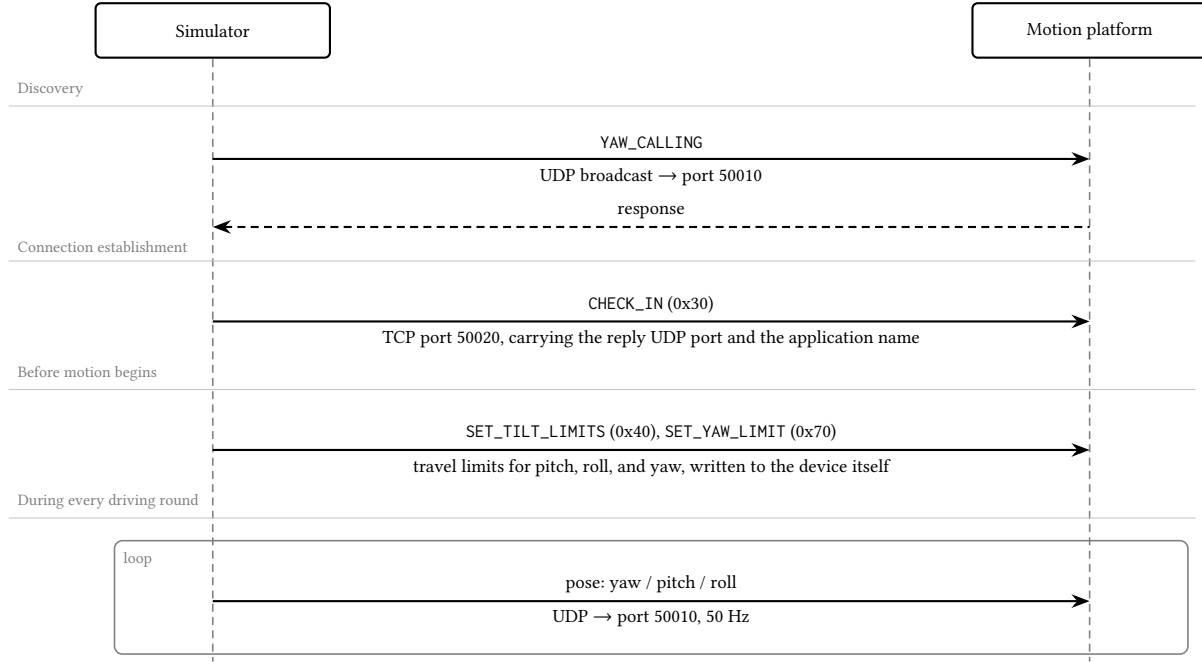


Figure 10: Connection and data flow between the simulator and motion platform. Discovery and the pose stream use UDP; connection establishment and travel-limit writing use TCP. Travel limits are written to the device before motion begins and are subsequently enforced by the device itself.



Figure 11: First-person view while driving (Draycott Place, Route R1): an ambient traffic vehicle ahead, a signal-controlled junction, the speed reading on the windscreen (19 km/h), route guidance on the instrument cluster (“20 m to finish”), and the live rearward view in the interior mirror.

behind it: five integer sliders (0–20, 21 tick marks), live numeric readouts, and a submit button disabled until every slider had been moved. Responses used a ray that preferred the Quest hand-tracking aim pose (right hand by default) and fell back to a Touch controller; confirmation was a pinch with hysteresis, or the trigger. The ray-canvas intersection was passed to Unity’s graphic raycaster, reusing standard pointer events so sliders could be dragged directly, and



Figure 12: Questionnaire presented in the virtual environment after each driving round. Participants answered five items with integer sliders from 0 to 20: mental demand, distraction from the road, informativeness about driving, pleasantness of use, and at-a-glance legibility. The endpoints were ‘0 = not at all’ and ‘20 = very much’. Each slider started at the midpoint of 10 and had to be moved at least once before submission.

the ray was drawn as a thin line so participants knew where they were pointing (Figure 12).

4.9 Route Generation

The road network was exported as a junction-edge graph, each edge storing length and a class-derived lane half-width. A depth-first search enumerated non-revisiting paths under an edge-count

bound; candidates had to satisfy a round-suitable total length, a minimum cumulative turning angle, and avoidance of a predefined excluded area. Eligible candidates were ranked and spatially deduplicated (one per 250 m grid cell, at least 30 m from any selected route); one block-encircling route came from a cycle search confirmed by a point-in-polygon test.

Route polylines lay on the road centreline with per-point lane half-widths, and the guidance arrow, driving surrogate, and route-completion test all read the same polyline; candidates were inspected via an editor-only overlay.

5 Results

This section reports, in research-question order, the multi-objective indicators across the fixed-order phases, the recommended configurations and same-route references, design differentiation, and feasibility and device-condition analyses. Every value was checked against the local analysis output; release status is described in the Open Science section.

5.1 Dataset

All 16 participants (P1–P16) completed all 21 rounds, giving 336 observations: 15 sampling rounds, five optimisation rounds, and one final-design round per participant. P1–P8 completed the protocol with the motion platform active. After failure of the platform's pitch axis, P9–P16 completed it with the seat stationary, producing two equally sized groups of eight. All completed sessions entered the analysis; none was excluded.

5.2 Within-Session Optimisation Progress (RQ1)

5.2.1 Hypervolume Progress. Between the end of sampling at Round 15 and the end of optimisation at Round 20, cumulative hypervolume increased for 15 of the 16 participants and remained unchanged for one; the median relative gain was 10.9%. The paired comparison was $V = 135$, $p < .001$ (exact $p = .00006$), $r = .71$, Cohen's $d = 0.86$, 95% CI [0.28, 1.43]. Because cumulative hypervolume increases monotonically and phase order was fixed, this describes the change in the cumulative front after five additional evaluations.

As a budget-matched internal sensitivity comparison, the hypervolume gain across the five optimisation rounds exceeded that across the immediately preceding five sampling rounds: $V = 107$, $p = .044$, $r = .36$, $d = 0.58$, 95% CI [0.04, 1.10]. The two five-round windows nevertheless occurred at different times in the session and were based on different search mechanisms and paths, so this remains an internal comparison under a fixed ordering (Figure 13).

5.2.2 Comparison of the Sampling and Optimisation Phases. The five objectives formed one outcome family; raw p values are kept in the table for auditability, and ART and Friedman results were separately Holm-corrected across the five objectives. After correction, the omnibus phase associations remained significant for the four subjective objectives but not for the simulated energy-intensity proxy. These omnibus tests do not identify which two phases differ and do not remove the fixed-order confound.

The omnibus phase test for task load gave ART $F(2, 30) = 10.34$, raw $p < .001$, partial $\eta^2 = .41$; the Friedman result was $\chi^2(2) =$

11.38, raw $p = .003$, Holm $p = .006$. Raw ART p values for informativeness, pleasantness, and legibility were all $< .001$ and remained $< .005$ after Holm correction; their Friedman Holm p values were $< .005$, $.006$, and $< .005$, respectively. The simulated energy-intensity proxy showed no omnibus phase association: ART $F(2, 30) = 2.37$, $p = .111$; Friedman $\chi^2(2) = 2.00$, Holm $p = .368$ (Table 6).

Round number, practice, fatigue, rating strategy, and model-driven selection all covaried with the fixed phase order, so Figure 14 and Figure 15 show round trajectories descriptively, not as causal effects.

Figure 15 shows the energy proxy with the same conventions: substantial round-to-round fluctuation, a non-significant omnibus test, and a descriptive trend only.

5.2.3 Pareto Hit Rate. To avoid averaging five competing objectives, I independently recomputed each participant's five-dimensional Pareto front across the first 20 rounds; moocore and emoa agreed on all 320 rounds and matched the log flags, and 93 rounds were Pareto-optimal within their participant (Table 7). The hit rate depends on all five observed objectives, including the four ratings.

Calculated per round, the Pareto hit rate was 0.24 in the sampling phase and 0.44 in the optimisation phase; the paired comparison gave $V = 111$, $p = .024$, $r = .40$. The difference is in the direction expected from model-driven selection, but the two phases were fixed in sequence and the front was defined retrospectively using all 20 rounds. It is therefore described as a fixed-order phase association, not interpreted as a causal doubling of the hit rate by the optimiser (Figure 16).

5.2.4 Final Recommended Design. Round 21 repeated a recommended configuration observed during the first 20 rounds. The selector chose from the observed Pareto candidates using normalised distance to the ideal point and predefined tie-breaking rules; eight of the 16 recommended configurations came from sampling and eight from optimisation. Relative to the same-route sampling baseline, the recommended configuration improved task load, informativeness, pleasantness, and legibility by 15.8, 19.8, 16.6, and 18.2 points, respectively, and all remained significant after Holm correction within the five-objective family. The difference in the simulated energy proxy was -0.4 kWh/100 km, $V = 62$, $p = .782$, Holm $p = .782$. Matching controlled route identity but not order, practice, traffic, or repeated-measurement conditions (Table 8).

The only comparison in which the selected design was inferior was against that participant's own best sampling round: the latter used 7.3 kWh/100 km less energy and was also slightly better on the subjective objectives. This is expected because the optimiser selected a compromise from across the front, whereas the 'best of 15 rounds' baseline is itself a maximum selected from a larger sample.

5.2.5 Measurement Properties of the Energy Objective. Energy was the only objective that did not move, so its measurement properties were characterised directly. Three quantities are reported as measurement results.

First, a post-hoc sensitivity analysis of the configured bounds. The energy proxy was declared to the optimiser as [0, 150] kWh/100 km, whereas observed values ranged from 16.0 to 64.6, occupying

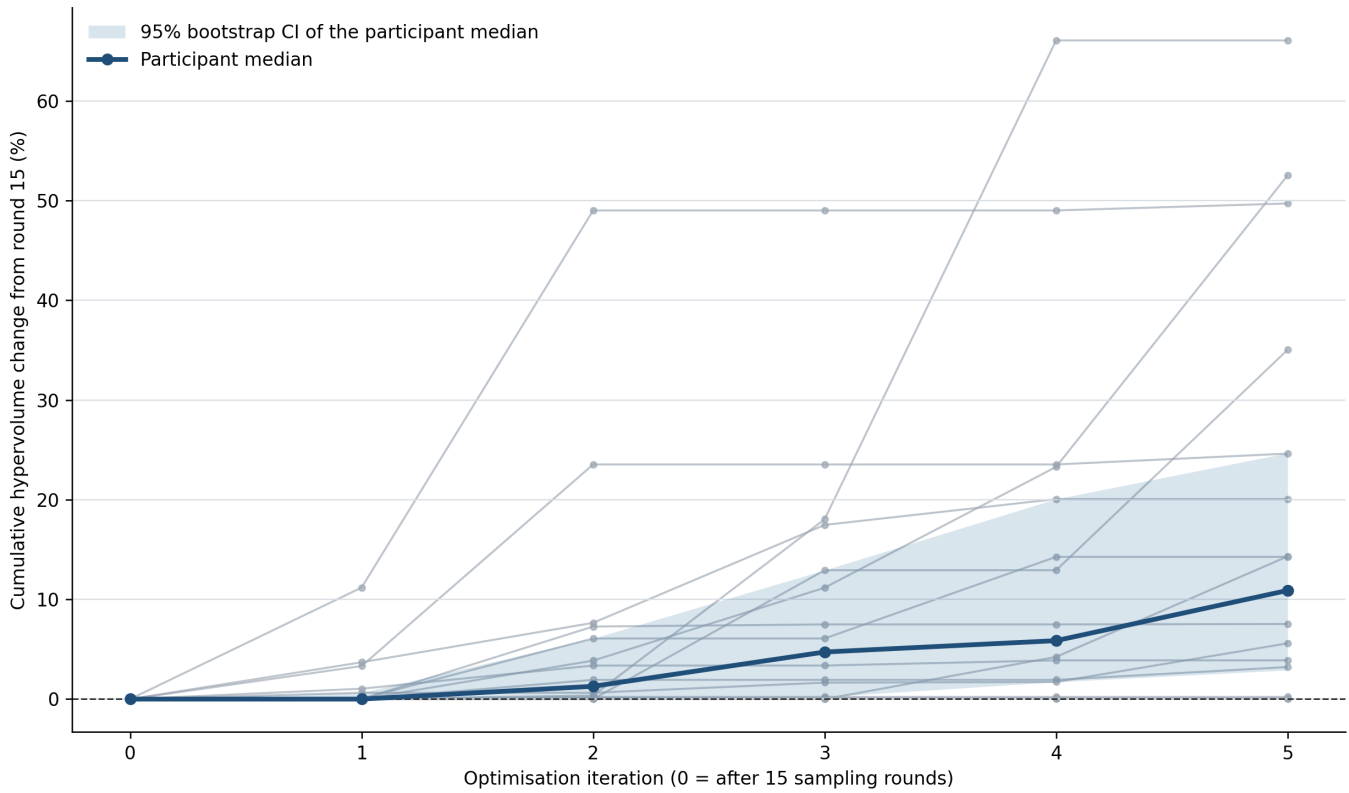


Figure 13: Percentage change in participants' cumulative hypervolume relative to its value at the end of sampling in Round 15. Thin lines represent individual participants; the thick line is the median at each round, and the shaded area is a participant-level bootstrap 95% confidence interval. The dashed line marks the five model-driven optimisation rounds. An increase in this cumulative metric should not, by itself, be interpreted as a causal optimisation effect.

Table 6: Omnibus associations between phase and the five objectives ($N = 16$; one value per participant per phase; phases = sampling, optimisation, and final design). The table reports raw ART and Friedman p values; the text separately reports the conclusions after Holm correction across five objectives for each test. The tests do not provide pairwise phase comparisons, and the fixed phase order also limits causal interpretation.

Objective	ART $F(2, 30)$	p	partial eta squared	Friedman chi-sq(2)	p
Simulated energy proxy (kWh/100 km)	2.37	.111	.14	2.00	.368
Task load	10.34	< .001	.41	11.38	.003
Informativeness	22.44	< .001	.60	16.62	< .001
Pleasantness	15.98	< .001	.52	12.88	.002
Legibility	22.84	< .001	.60	13.88	< .001

only 32.4% of that interval, with a normalised standard deviation of 0.060. When the bounds were changed post hoc to [15, 65], the internal five-round-window comparison changed from $V = 107$, $p = .044$, $d = 0.58$ to $V = 117$, $p = .009$, $d = 0.80$ (Table 9). Because the new bounds were determined from the same data, this result shows only that the conclusion was sensitive to scaling; it is not independent evidence of a validated performance improvement or of how alternative bounds should be selected.

Second, cross-round retest differences for the same configuration. Round 21 repeated a previously presented configuration, but retesting also changed the point in the session and could change

route state, traffic interaction, and driving familiarity. Mean energy-proxy values in the source and retest rounds were 29.318 and 34.382 kWh/100 km, respectively, and the paired difference was significant (source minus retest = -5.064, $p = .00336$). This difference therefore cannot be called pure 'measurement noise'. It combines variation from session timing, route or traffic, driving behaviour, and error in the proxy model, and it also shows that the energy proxy from one short-route round was unstable.

Third, variance-association models showed that, in the fitted within-participant models, the proportion of harsh acceleration and mean absolute acceleration were associated with the energy

Table 7: Pareto-optimal rounds by participant. A round was Pareto-optimal when no other round for the same participant was no worse on all five objectives and strictly better on at least one. Fronts were calculated independently with moocore and emoa, which agreed on all 320 rounds.

P	Pareto rounds (of 20)	In sampling (of 15)	In optimisation (of 5)	Rate sampling	Rate optim.
P01	4	4	0	0.27	0.00
P02	9	6	3	0.40	0.60
P03	8	7	1	0.47	0.20
P04	5	2	3	0.13	0.60
P05	7	6	1	0.40	0.20
P06	3	0	3	0.00	0.60
P07	7	4	3	0.27	0.60
P08	9	7	2	0.47	0.40
P09	9	5	4	0.33	0.80
P10	2	1	1	0.07	0.20
P11	2	1	1	0.07	0.20
P12	5	4	1	0.27	0.20
P13	7	4	3	0.27	0.60
P14	8	5	3	0.33	0.60
P15	5	1	4	0.07	0.80
P16	3	1	2	0.07	0.40
All	93	58	35	0.24	0.44

Table 8: Comparison of the recommended configuration in Round 21 with two references ($N = 16$; exact two-sided Wilcoxon signed-rank tests; d is paired-samples Cohen’s d ; Holm correction was applied across five objectives within each reference family). The same-route sampling baseline matched route identity only. The best sampling round takes the best value from the 15 rounds separately for each objective and therefore does not represent one common design.

Objective	Dir.	Final M	Matched-route M	Delta	p	Holm p	d	vs best sampling round
Simulated energy proxy (kWh/100 km)	min	34.4	34.0	-0.4	.782	.782	-0.08	worse by 7.3, Holm $p = .001$
Task load	min	20.3	36.1	15.8	.009	.018	0.84	n.s. (Holm $p = 1.00$)
Informativeness	max	85.3	65.5	19.8	< .001	< .001	1.15	n.s. (Holm $p = 1.00$)
Pleasantness	max	83.8	67.1	16.6	.004	.013	0.85	n.s. (Holm $p = 1.00$)
Legibility	max	84.4	66.2	18.2	< .001	.003	1.20	n.s. (Holm $p = 1.00$)

Table 9: Occupancy of the configured range by each objective and the resulting leverage in the normalised space used by the optimiser.

Objective	Configured bounds	Observed min	Observed max	Box used (%)	Normalised SD
Simulated energy proxy (kWh/100 km)	0 to 150	16.0	64.6	32.4	0.060
Task load	0 to 100	0	90.0	90.0	0.192
Informativeness	0 to 100	0	100.0	100.0	0.204
Pleasantness	0 to 100	5	100.0	95.0	0.212
Legibility	0 to 100	0	100.0	100.0	0.213

proxy, whereas mean speed had only a weak association. Driving-behaviour covariates corresponded to a model increment of approximately 59.1%, and the seven display parameters to approximately 5.1%. In a model restricted to sampling rounds, the incremental test of display parameters against the energy residual was $\chi^2(7) = 13.6$, $p = .059$. These percentages are associative or model-attributed shares under a particular specification, not causal proportions by which the variables ‘explain’ real energy use, and they do not establish either that the display is unrelated to energy or the opposite.

5.3 Differentiation of Personalised Designs (RQ2)

5.3.1 Parameter Values of the Recommended Configurations. Across the recommended configurations in Round 21, six of the seven dimensions covered almost the full normalised range. The numerical-score dimension was narrower ($M = 0.37$, $SD = 0.21$,

maximum = 0.77). These values describe configurations selected from observed candidates; they should not be called converged or stable individual optima, because each participant provided only 20 modelling evaluations and one retest (Figure 17, Table 10).

5.3.2 Which Elements Were Retained. In the final designs, the leaf gauge was retained for 11 of the 16 participants, while the acceleration icon and vehicle markers were each retained for eight, the framing element for seven, and the numerical score for five. No optimiser-controlled element was retained by everyone, and none was removed by everyone. The speedometer could not be suppressed by design and is excluded from these counts (Figure 18).

5.3.3 Are Between-Participant Differences Larger Than Variation Within the Optimisation Rounds? For every parameter, the between-participant standard deviation of the recommended values exceeded

Table 10: Parameters of the recommended configurations for the 16 participants: between-participant dispersion, search variation within the five optimisation rounds, and retention in the Round 21 configuration. The between-/within-participant decomposition is descriptive and does not attribute all differences to stable preferences.

Parameter	M	SD	Range	Within-SD	Ratio	Retained
Leaf gauge	0.56	0.31	1.00	0.17	1.82	11/16
Numeric score	0.37	0.21	0.70	0.16	1.34	5/16
Framing element	0.45	0.29	0.97	0.18	1.64	7/16
Speedometer	0.48	0.33	0.92	0.19	1.73	cannot hide
Acceleration pictogram	0.52	0.27	0.94	0.17	1.53	8/16
Vehicle markers	0.51	0.32	0.96	0.17	1.89	8/16
Opacity	0.43	0.32	1.00	0.17	1.82	n/a

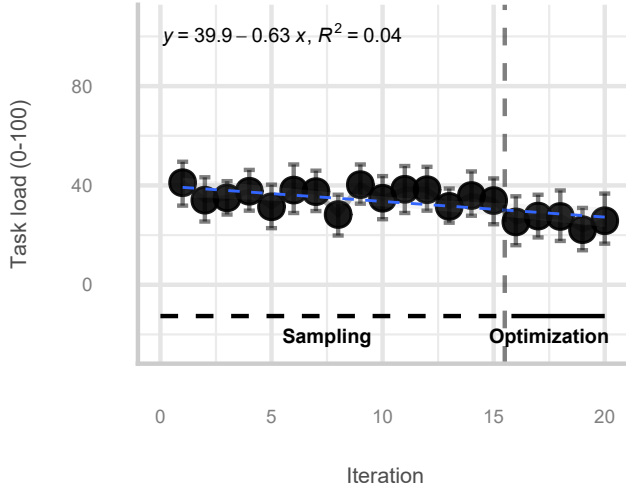


Figure 14: Task load across Rounds 1–20. Filled points with error bars show the participant mean and 95% t confidence interval for each round; the solid line is a descriptive linear trend across all 20 rounds, and the vertical dashed line marks the transition from sampling to optimisation after Round 15. The final repeat drive in Round 21 is excluded. The trend cannot distinguish practice, fatigue, rating drift, and the selection mechanism.

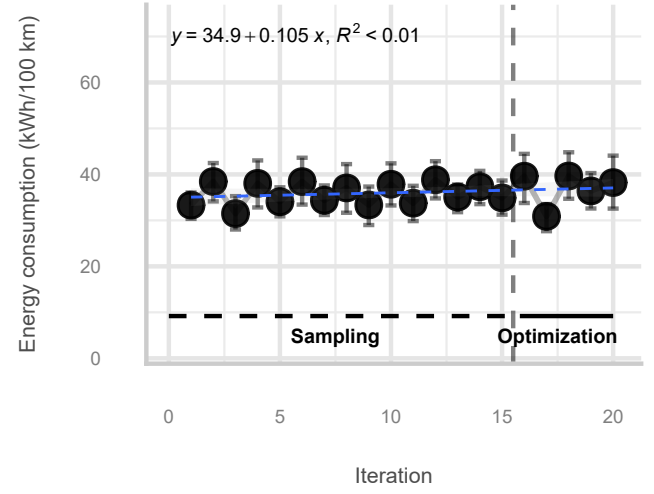


Figure 15: Simulated energy-intensity proxy across Rounds 1–20. Filled points with error bars show the participant mean and 95% t confidence interval for each round; the solid line is a descriptive linear trend across all 20 rounds, and the vertical dashed line marks the transition from sampling to optimisation after Round 15. The final repeat drive in Round 21 is excluded.

the mean within-participant standard deviation across the five optimisation rounds, with ratios of 1.34–1.89. This indicates participant-associated differentiation in the selected configurations, but the comparison also contains variation caused by different initial samples, search paths, and the limited budget; it cannot by itself establish stable individual preferences.

A label-permutation test further showed that the 93 Pareto designs were closer to others carrying their original participant label than under random relabelling (observed separation +0.090, 5,000-permutation $p = .0002$). The procedure tested association with the label but did not reconstruct each Sobol seed and adaptive search path under permutation, so participant preference remains inseparable from path differences. In the random-slope models, the associations of the framing element with pleasantness and task load showed between-participant heterogeneity, both with Holm-corrected $p < .001$; no other parameter-objective combination passed correction.

Open-ended responses also included opposing evaluations of the framing element: four participants described it as annoying or distracting, while another listed it as the favourite element. This is consistent with the heterogeneity models but remains participant-associated evidence from a small sample and requires retesting across days to confirm stability.

5.3.4 Is a Participant's Own Design Better for That Participant?

Determining whether a person's configuration is better than other people's configurations requires predicting responses to designs they have never experienced. The median leave-one-round-out R^2 of the per-participant ridge-regression response surfaces was -0.067, and only 7/16 values were above zero; reducing the model to three parameters did not improve performance (-0.081). A negative value means that the model predicted the held-out round no better than the participant's own mean, so the subsequent transfer results are not confirmatory.

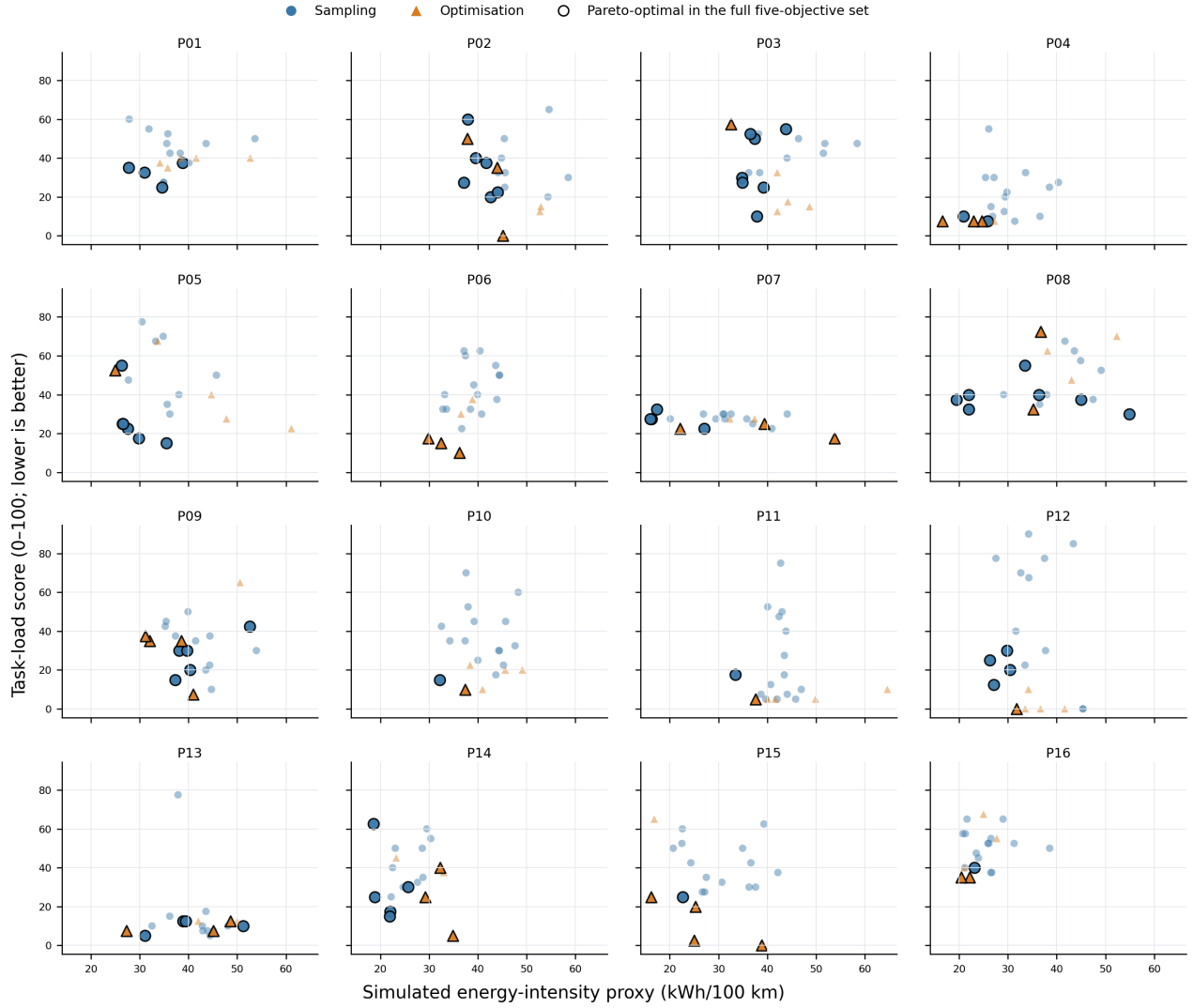


Figure 16: Two-dimensional projection of each participant’s designs from the first 20 rounds onto simulated energy-intensity proxy and task load. Sampling and optimisation rounds use different symbols; colour highlights points classified as Pareto-optimal using all five objectives. Axes are fixed across panels. Occlusion and relative position in the two-dimensional projection are not equivalent to five-dimensional dominance, and no connecting lines imply a trajectory.

Given this limitation, the transfer analysis is presented only as an exploratory result. On average, the models assigned higher predicted scores to participants’ own designs than to other participants’ designs and to the common design selected by the group model. However, each participant’s own design was also closer to their training data, and the underlying response surfaces performed poorly under cross-validation; adding a distance covariate did not remove the dependence on the model. These values can therefore generate hypotheses only and do not establish a transferable advantage of a personalised design for its owner.

A related property of the objective set affects how hypervolume should be interpreted. Within participants, the five objectives were correlated: task load and the three acceptance constructs loaded together on the first principal component, accounting for 50.7% of variance; energy loaded almost alone on the second, accounting for 20.1%; and only the first component survived a parallel analysis against column-wise shuffled data. The effective dimensionality was 3.02 out of five. Energy was the only objective that was not another facet of the acceptance construct, and its within-participant correlations with the other four objectives ranged from -.01 to .04.

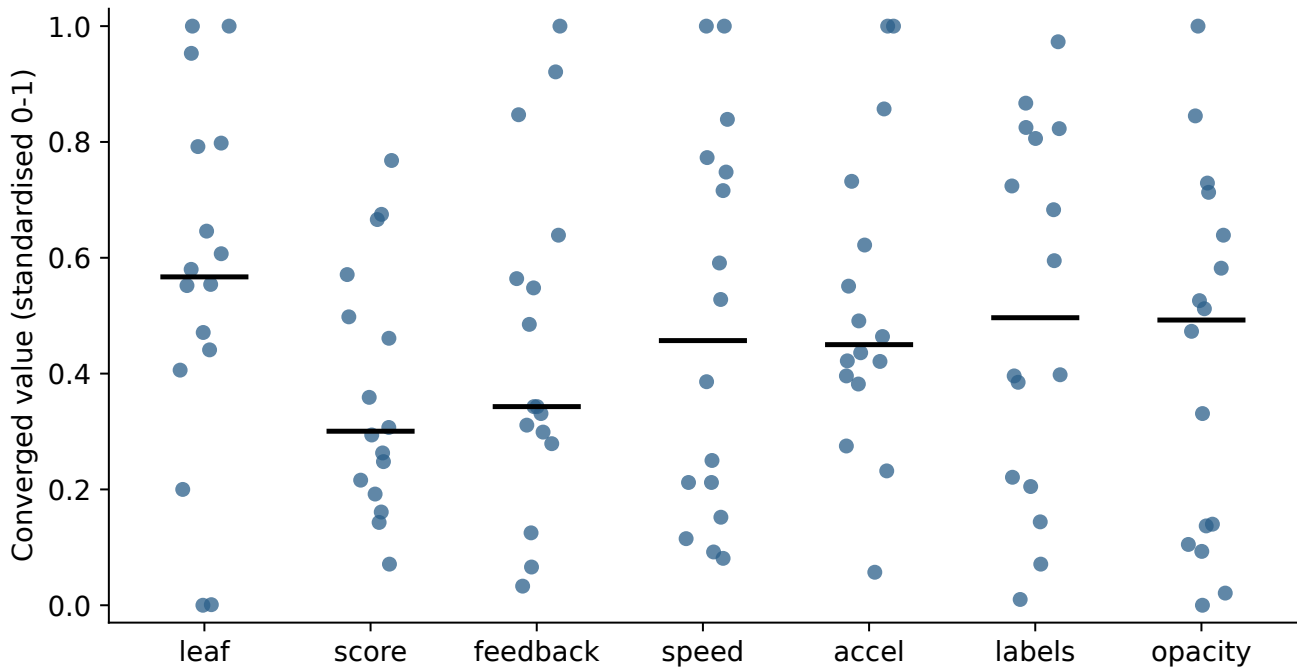


Figure 17: Values of the seven design parameters in the recommended Round 21 configurations. Each marker represents one participant; solid horizontal lines show medians.

5.4 Feasibility, Acceptability, and Comparison of Motion Conditions (RQ3)

5.4.1 Session Duration and Completion. All 16 participants completed all 21 rounds. The median duration of the in-simulator portion was 24.2 minutes, with a range of 16.5–38.0 minutes. Longer rounds occurred mainly early in the session and were often accompanied by very low speeds and repeated contacts, which is consistent with participants becoming familiar with the steering wheel, pedals, and simulator vehicle. The 'collision count' in the logs comprises collision-entry events recorded by CollisionLogger: entries for the same collider were rate-limited within two seconds, normal rolling contact with the terrain was excluded, but another collision after the cooldown could be counted separately. It is neither a count of physical contact frames nor a basis for diagnosing a system fault. Sessions recorded 12–127 events (median 56). The maximum energy-proxy value in the formal experiment was 64.61 and did not reach the upper bound of 150.

5.4.2 Comparison of Motion and No-Motion Conditions. Because the motion condition changed over time after the eighth participant, the following analysis is exploratory only. Median hypervolume gains were 4.40% in the motion group and 14.29% in the stationary group ($U = 24$, $p = .442$, rank-biserial correlation = $-.25$, bootstrap 95% CI for the between-group median difference $[-27.46, 31.09]$). Median changes in Pareto hit rate were .0667 and .300, respectively ($U = 21$, $p = .279$, $r = -.344$, 95% CI $[-.6, .2]$). The p values for the five final same-route changes ranged from .328 to .959. After Holm correction across the seven sensitivity tests, all adjusted p values

were 1.000, and all confidence intervals for differences crossed zero. With only eight participants per group, non-random conditions, and complete overlap with recruitment order, the results neither show a clear device effect nor establish equivalence between the conditions.

5.4.3 Post-Study Questionnaire. All eight participants in the motion group rated the seat movement as 'just right'.

On the five-point post-study scales, 'the final design suited me' had a mean of 4.38 ($SD = 0.72$), 'the display changed towards my preferences across rounds' 4.12 ($SD = 0.81$), 'I would use such a display in a real car' 3.94 ($SD = 1.12$), and perceived distraction 2.00 ($SD = 1.21$). The item 'the final display helped me drive more efficiently' had a mean of 4.31 ($SD = 1.08$), but participants' judgements were influenced by the visible eco-score feedback in the interface and should therefore be interpreted as self-reported perceived helpfulness. Mean motion discomfort was 2.69 ($SD = 1.45$), and five participants gave a rating of 4 or 5.

Open-ended responses mirrored the parameter results, including their differentiation. Eight participants listed the leaf gauge as their favourite element, and one described an ideal display containing only a single leaf. The framing element received the most criticism: four participants called it annoying or distracting, and one said that the numerical score changed too quickly and caused dizziness when watched. The same elements nevertheless had advocates. One participant in the no-motion group preferred the advice text because it explained how to improve; another requested that the vehicle markers be retained; and another praised the colour-changing

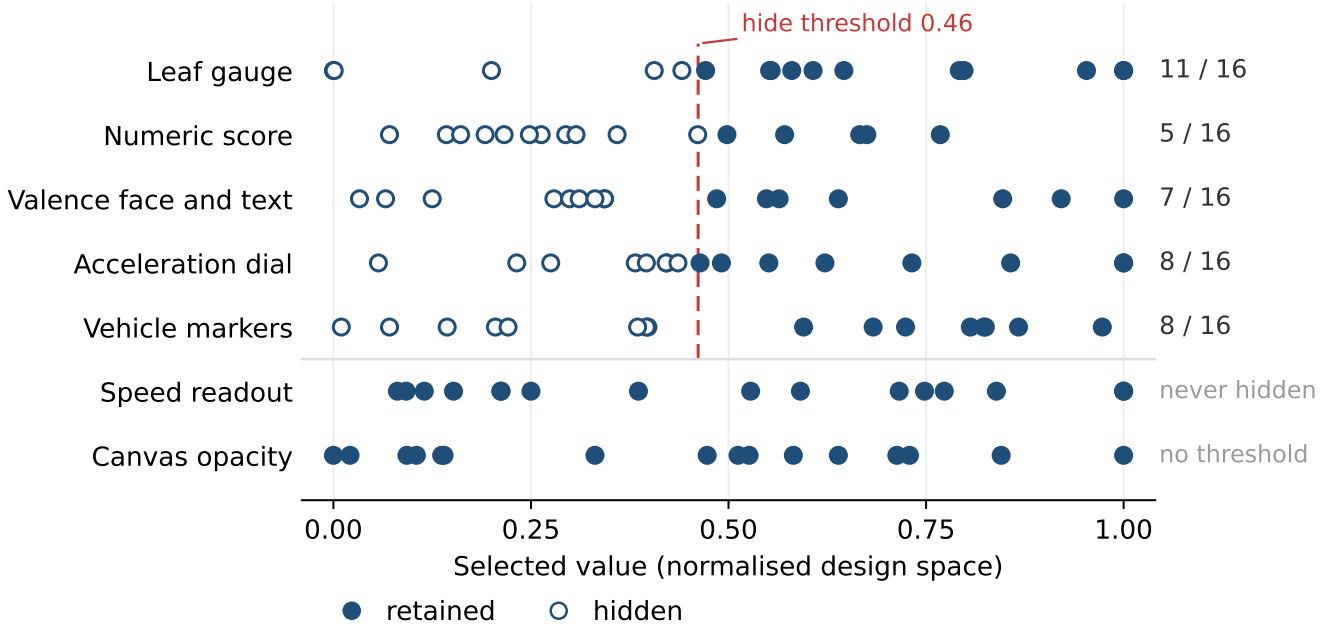


Figure 18: Normalised parameters of the recommended Round 21 configurations for the 16 participants. Filled points indicate that an element was shown and open points that it was hidden. The dashed line is the normalised hiding threshold of $0.6/1.3 \approx 0.462$, derived from the physical scale threshold in the implementation. The speedometer cannot be hidden and opacity does not use this threshold; both are shown below the separator.

speedometer. Twelve of the 16 participants said that the final design was, or was close to, their ideal design.

5.5 Summary of Results

Overall, the fixed-order experiment showed that cumulative hypervolume increased between Rounds 15 and 20 for 15/16 participants, and that both the five-round-window comparison and Pareto hit rate moved in the direction favourable to the model-driven phase. The four subjective outcomes were associated with phase, whereas the simulated energy proxy showed no significant phase difference. The observed configuration selected for Round 21 improved subjective ratings relative to the same-route sampling reference, although this comparison remained affected by order, practice, and rating drift. Configurations showed participant-associated differentiation, while the transfer analysis was exploratory because its cross-validated R^2 was negative. All participants completed the protocol; the device-condition sensitivity analysis was inconclusive.

6 Discussion

6.1 What Within-Session Optimisation Can and Cannot Establish

Each participant provided only 20 modelling evaluations for a seven-dimensional, five-objective search: 15 for space filling and five for model-driven selection. This budget yields only a sparse Pareto approximation and cannot establish convergence, so RQ1 describes a specific experimental workflow; the independent effect of the

optimiser relative to random search, a fixed default display, or practice lies beyond this design.

The five model-driven iterations came at the end of the session, so any change in hypervolume, ratings, or behaviour was simultaneously exposed to practice, fatigue, changes in rating strategy, and the route cycle. Drift in the feedback users give over time is a recognised concern in interactive machine-learning systems [10].

Cumulative hypervolume increased for 15/16 participants, with a median gain of 10.9%, and the gain over the five optimisation rounds exceeded that over the preceding five sampling rounds ($V = 107$, $p = .044$). Because cumulative hypervolume cannot decrease by construction [8] and the two windows occurred in a fixed sequence, these values only show that the workflow continued to expand the observed front within a limited budget.

The rise in Pareto hit rate from 0.24 during sampling to 0.44 during optimisation is likewise consistent with model-driven selection. However, the front was defined retrospectively from all 20 rounds, four objectives came from ratings, and phase order was fixed, so the hit rate does not remove order effects or rating drift.

The four subjective outcomes were more favourable later in the session, whereas the simulated energy proxy showed no significant phase change, consistent with the noted difficulty of obtaining realistic energy signals from a driving simulator [3]. The most conservative reading is that ratings entered the loop directly and could shift quickly, whereas the energy proxy followed a longer measurement chain through driving behaviour and short-route

extrapolation. Neither the source of the subjective change nor the null energy result generalises to real vehicles.

Changing the energy bounds post hoc from [0, 150] to [15, 65] strengthened the internal comparison, showing that acquisition was sensitive to objective scaling. Because the new bounds came from the same data, this is a sensitivity analysis; replications should set bounds from an independent pilot or a preregistered rule.

The energy proxy also differed by approximately 5.06 kWh/100 km between Round 21 and its source round, which cannot be reduced to measurement noise because the two drives occurred at different times, under possibly different traffic and familiarity. Separating measurement error from session-level change requires longer routes, repeated retests, and randomised phase and route.

Association models attributed more energy variation to driving-behaviour covariates than to display parameters, but these shares depend on model specification and are associative only: the display may act through behaviour, and route, traffic, and temporal factors may be omitted. The study therefore does not use them to establish either that the display is unrelated to energy or the opposite.

The conclusion for RQ1 is limited to this fixed-order workflow: subjective and Pareto signals consistent with model-driven selection emerged within a realistic budget. Without an independent non-adaptive condition, the study cannot determine whether the method outperforms a carefully designed fixed display; the most direct follow-up randomly interleaves optimisation with matched non-adaptive selections and includes a strong fixed default.

6.2 Why Personalised Displays Differed

The RQ2 observation that does not depend on any average phase improvement is the differentiation of recommended configurations and Pareto sets under participant labels. Each participant also received different initial Sobol points and followed a different adaptive path, so these differences combine preference, seed, path dependence, and finite-sample error; whether they are stable individual preferences awaits retesting across days.

Six of the seven dimensions covered nearly their full ranges, between-participant dispersion exceeded variation within the five optimisation rounds, and label permutation showed Pareto sets clustering by participant. The permutation kept the observed points but did not rerun an identical search path per label, so it cannot separate 'who provided the ratings' from 'which candidates were seen'; the random-slope results and opposing open-ended responses add complementary evidence, but the sample is small.

'Different designs' and 'a person's own design is better for that person' are distinct claims. The transfer analysis had a median leave-one-round-out R^2 of -0.067 , with only 7/16 values positive, so individual response surfaces predicted unseen designs poorly and the transfer rankings and personalisation gain can only generate hypotheses. Confirmation requires optimisation repeated across days, participants experiencing others' or a common configuration, and hierarchical or multi-task models.

Three explanations from previous work fit this pattern. The first concerns use: Stillwater and Kurani [20] found instantaneous performance feedback used mainly to experiment with and learn new behaviours, and average-performance feedback used mainly to set and achieve goals. My numerical score was instantaneous,

so disagreement about it may track whether a driver still needs a quantity for experimentation.

The second concerns attention, with a caution: Jamson et al. [12] found that a more informative visual display reduced control error and that an added auditory channel reduced visual distraction, so attentional cost depends on how information is distributed across sensory channels rather than on amount. My display was purely visual, so opacity directly set the visual resource it occupied, and drivers differed in how much they would allocate; this difference is an observation of the present study without a counterpart conclusion in the literature. The third concerns representation: Dahlinger et al. [7] showed that numerical and symbolic presentations of the same information can produce different behaviour, so preferences between a score and a gauge are more than taste.

A fourth explanation is explicitly speculation. The leaf gauge, numerical score, and framing element partly report the same underlying eco-score, so an optimiser minimising task load while maximising informativeness has an incentive to retain only one of them, and which one may hinge on small early-rating differences. Part of the differentiation would then be a property of the design space rather than of the driver; distinguishing the two requires re-optimising with the same participant on another day, which I did not do.

Whichever explanation applies, the practical implication matches the conclusions reached from different directions by Brouwer et al. [3] and Sumner et al. [21]: fixing one display for all drivers removes a genuine source of variance.

Open-ended responses made the differentiation concrete: eight participants chose the leaf gauge as their favourite (one wanted only a single leaf), four called the framing element annoying or distracting, yet one preferred it precisely for its advice text, one asked to keep the vehicle markers, and one praised the colour-changing speedometer. One driver removed an element that another would use the display specifically to obtain.

6.3 Designing for a Constrained Evaluation Budget

Two design decisions protected the evaluation budget and are transferable. First, element presence was folded into element size, keeping the search space continuous: a 14-dimensional mixed discrete-continuous space would be far harder to search in 20 evaluations, and modelling it would add assumptions a small sample cannot test. The cost is that the optimiser cannot distinguish an element intentionally removed from one made very small; Dudley and Kristensson [10] note that such representational choices shape what an interactive system can learn.

Second, acceptance stayed split into three objectives. Averaging them, as Jansen et al. [14] did with two acceptance items, would ease optimisation, but informativeness, pleasantness, and legibility could move in opposite directions after the same design change, and averaging would conceal exactly the conflict that motivates the optimiser. Whether this trade-off was right is open, because no averaged-objective variant was run.

The protocol itself - 21 rounds in a head-mounted display - approaches the upper end of a reasonable range: per-person budgets of 15-40 evaluations are typical for the method [4, 14], but unusual

for a driving-simulator study. There is also an experiential cost: Chan et al. [4] found that optimiser-guided participants reported lower agency and expressiveness despite better coverage and solutions, and participants here had no control over which design appeared next, so that loss of agency was present in a stronger form.

6.4 Implications for Eco-Driving Display Design

If the observed differentiation is real, the design question shifts from ‘which single display should leave the factory?’ to ‘which parts should be adjustable, and who adjusts them?’. The seven dimensions answer this concretely: parameters on which participants converged are candidates for fixed design decisions, and parameters on which they disagreed are candidates for user control or adaptation.

Vehicle markers, the leaf gauge, and overall opacity showed the greatest differentiation (ratios of 1.82–1.89); the numerical score showed the least (1.34) and was retained in only five of the 16 final designs. Whether eco-information is expressed numerically or symbolically has itself been shown to change fuel consumption in the field [7], so low retention of the numeric channel is a substantive design signal.

This conclusion is more restrained than ‘complete per-driver optimisation’ and more deployable: a production system cannot require 20 instrumented drives with a Bayesian optimiser, but it can expose two or three settings, and a study of this kind identifies which. This supports, from another direction, the position of Tu et al. [23]: adoption of eco-driving guidance depends on whether it fits the driver.

A further implication concerns what the display is for. I operationalised the usefulness dimension of acceptance as informativeness because fuel consumption was imperceptible in the simulator and behaviour change could not be rated immediately; the acceptance objective therefore measured how well the display informed, and persuasiveness was not measured. A display optimised to inform is not necessarily one optimised to change behaviour: Sanguinetti et al. [17] likewise map information, timing, and presentation to attention, motivation, and learning, separating what a display communicates from what behaviour it produces.

6.5 Ethics and UK AI Governance Principles

This section applies the five cross-sector AI regulatory principles proposed by the UK Government [9], which emphasise understanding risk in context, to the system’s research controls, residual risks, and deployment requirements.

6.5.1 Safety, Security, and Robustness. The virtual environment removed real-road collision risk but created risks of virtual-reality discomfort, visual-vestibular conflict, and mechanical hazards from the motion platform; the information sheet and risk assessment specified stopping, resting, and limiting device travel. Route guidance was excluded from the design space and the speed display kept minimum size and readability levels as design constraints. The platform failure during the study shows that robustness is not captured by mean frame rate; deployment would additionally require fault detection, safe fallback, latency measurement, and on-road validation.

6.5.2 Appropriate Transparency and Explainability. Participants were told the display would be adjusted according to their ratings, but not the surrogate model, acquisition function, or generation of the next configuration. That disclosure sufficed to signal adaptation, yet may have produced expectation effects and would not support informed use in deployment, where a driver-facing version should explain what is collected, what is optimised, why the configuration changed, and its uncertainty, without adding distraction while driving.

6.5.3 Fairness. Personalisation can be fair only if the design space and training data cover different needs. The sample was small (16), concentrated in age, and excluded several health conditions, so differences by age, disability, driving experience, or culture cannot be assessed, and size bounds and hiding thresholds exclude needs outside those boundaries. Future work should predefine accessibility constraints, report subgroup performance, and avoid imposing longer cold starts or less stable configurations on sparsely represented groups.

6.5.4 Accountability and Governance. A per-participant system requires configuration-level logs to reconstruct what someone saw in a given round, which data and model version produced it, and why it was selected. This study stored the seven-dimensional parameters and five objectives, with public release constrained by the consent materials and re-identification risk. A deployed system would further need a named data controller, approval and incident-review processes, version rollback, and ultimate human responsibility; the optimisation tool itself cannot be the accountable party.

6.5.5 Contestability and Redress. Participants could stop the complete study at any time, but had no round-level control to skip the current configuration, return to the previous one, or reset the personalisation model; ratings affected only subsequent rounds. This was methodologically motivated but should not transfer to a product, which should let users leave personalisation, pause learning, restore a safe default, withdraw data, and reach human support, without reducing the availability of basic driving information.

6.5.6 Adaptivity and Autonomy of the System. The system refitted its model after each round using new ratings and the energy proxy, demonstrating adaptivity, and proposed the next configuration without per-round researcher input, demonstrating limited autonomy bounded by the predefined seven-dimensional limits and five objectives. Ultimate research responsibility remained with the researchers; assessing risk in context is more meaningful than assigning the system a single categorical label.

6.5.7 Privacy, Social Harm, and Misuse. Per-participant optimisation continuously collected driving trajectories and subjective ratings, which may permit re-identification through behavioural patterns even after names are removed, and the consent materials contain an ambiguity about the scope of public data, so high-dimensional trajectories must not be released in advance. Foreseeable misuse includes applying the same loop to prolong glances, increase advertising engagement, or optimise other objectives against drivers’ interests. Mitigations include data minimisation, purpose limitation, independent safety constraints, access controls, and a re-identification assessment before release.

6.6 Limitations

The most important limitation is the absence of a non-adaptive baseline. 'Baseline' throughout refers to the internal, route-matched comparison with the participant's own sampling rounds, not the fixed, carefully designed default display a manufacturer might deploy; no such condition was included, and all claims in the thesis are limited accordingly.

The measurement instruments were simplified: two items replaced the six subscales of NASA-TLX [11], and three brief items replaced the complete acceptance scale [24]. Single items cannot provide internal-consistency estimates, so the results apply to the questions asked and cannot be generalised to the validated constructs they were adapted from; Jansen et al. [14] made the same shortcut, which is conventional in this class of study but not cost-free.

A more fundamental limitation is the simulator's ecological validity. Harsh acceleration and speeding had no real consequences, and fuel consumption, imperceptible here, is perceptible in a real vehicle. One result makes this concrete: Pietra et al. [16] found in a ten-participant virtual-reality simulator that haptic feedback reduced the mean power index by 14%, the combined condition reduced it by 11%, and visual feedback alone increased it by 11%. The present display was purely visual and evaluated in virtual reality, so the closest published comparison points in an unfavourable direction, and I cannot exclude the same effect on the energy objective.

Performance logs recorded approximately 73,245 frames over 16.96 minutes of partially overlapping periods, with a mean of 71.99 fps - close to the 72 Hz ceiling, but not an end-to-end motion-to-photon measure and not covering every session. The high discomfort reported by five participants therefore cannot be attributed to, or cleared of, any single cause among dropped frames, latency, virtual-reality presentation, motion cues, or visual-vestibular conflict.

Route and phase were not fully separated: the four routes cycled in a fixed order, the most difficult route appeared twice in the five optimisation rounds and three times in the 15 sampling rounds, and the final recommended configuration was always retested on the same route. The available evidence does not justify describing that route as the 'slowest' or as having 'the most junctions'; randomising or balancing route order would reduce this confound.

The same design carries order effects: each participant underwent optimisation once, within one sitting, with the sequence dependent on their previous ratings, so familiarity with the road network, the vehicle, and the rating task increased monotonically and was confounded with the sampling-to-optimisation transition. This is inherent to a single-session human-in-the-loop protocol.

Rating behaviour changed with round number: questionnaires were completed faster, extreme ratings became more frequent, and some relationships between items changed. Because phase was defined by round, these changes cannot be separated from model-driven selection; Pareto hit rate does not average ratings, but its dominance judgement still uses four rating objectives directly, so it is not independent of rating behaviour. Interleaving sampling and optimisation rounds and retesting across days would provide a stronger design.

The objective set had lower effective dimensionality than hypervolume assumes: task load and the three acceptance constructs loaded on one component holding half the variance, energy loaded almost alone on a second, and only the first survived parallel analysis, giving an effective dimensionality of 3.02 out of five. Hypervolume was nevertheless computed on all five axes, crediting dimensions with little independent information, so its within-participant change is more meaningful than its absolute value. The implication is two-sided: it supports combining the four subjective objectives in a replication, and it supports retaining energy as the only objective that is not another facet of acceptance.

The sample was concentrated in age with limited driving and simulator experience, restricting generalisation, and the equipment change after the eighth participant completely overlapped with temporal order. All seven Holm-adjusted *p* values in the exploratory sensitivity analysis were 1.000, with wide confidence intervals crossing zero, leaving the equipment effect uncertain.

7 Conclusion

This study explored eco-driving display design as a per-participant, human-in-the-loop multi-objective search problem. Across seven continuous display parameters, the system used five objectives: an energy-intensity proxy calculated from simulated road load, task load, and three acceptance ratings. Each participant completed 15 Sobol sampling rounds, five model-driven optimisation rounds, and one retest of an observed recommended configuration.

In a single session with sampling fixed before optimisation, cumulative hypervolume increased for 15/16 participants and the Pareto hit rate was higher during optimisation than during sampling. The four subjective outcomes were associated with phase, whereas the simulated energy proxy showed no significant phase difference. Recommended configurations improved subjective ratings relative to the same-route sampling reference and differed markedly between participants. However, phase order, practice, rating drift, different Sobol seeds, and search paths jointly limit interpretation. The present results cannot establish that the optimiser caused the improvement, that stable individual optima converged, or that energy use would decrease in real vehicles.

Beyond these associations, the study contributes an inspectable measurement chain - a driveable reproduction of a real urban district, a continuous display design space, a single-session optimisation loop, and logging that links every rating to the configuration and trajectory that produced it. Participants differed most in which elements were retained at all, and several recommended designs kept only a subset of the channels; subject to the caveat that search paths contributed, this points to element presence, rather than fine size tuning, as the first-order personalisation decision.

The most direct follow-up should randomly interleave model-driven and non-adaptive selections, include a strong fixed default, retest the same participants on different days, and validate behavioural effects on longer routes or real vehicles. The released code, with the data and analysis materials whose release is pending ethics review, is intended to make that comparison straightforward to build.

Open Science and Materials Availability

The public repository at <https://github.com/ArcadiaE/Sustainable-Personalised-Driving> provides the self-authored Unity/Python source code, the releasable pseudonymous study data, the R analysis scripts with their outputs, this dissertation, and documentation. Its README states that Cesium, CityGen3D, and other third-party assets are not distributed with the repository. This manuscript does not claim that 'all data and the complete project are openly available'.

The released set comprises the per-round master table, questionnaire-derived measures, per-participant optimiser observation and hypervolume logs, study configuration with per-file script hash manifests, the analysis scripts, and a data dictionary; this is the minimum required to reproduce the main statistical results. Raw post-session questionnaire exports (demographic combinations and free text), continuous high-frequency driving trajectories, detailed console logs, and the verbatim per-session script snapshots are withheld pending a re-identification-risk assessment. If the finally approved scope permits access only by authenticated researchers, the data will be supplied under that governance condition rather than misdescribing pseudonymisation as anonymisation.

Before submission of the final English manuscript, a version tag and commit hash will be frozen for the code and approved analysis materials. The record will include versions of Unity, Python/R, BoTorch, and key dependencies, random seeds, execution commands, licences, and the list of excluded third-party assets. An immutable archive and DOI will be created through Zenodo or the Open Science Framework. No DOI has yet been issued, so no invented identifier is entered here.

Generative AI Use Statement

This study used Anthropic Claude and OpenAI Codex as assistive tools. Their uses included learning and configuring virtual-reality equipment, the motion platform, and other hardware, as well as Unity and other software environments. They also assisted with developing, debugging, and checking simulator and analysis code; literature searching and verification of reference metadata; scripts for generating statistical figures; and proofreading, terminology harmonisation, and language polishing of the thesis. During preparation of the final English LaTeX manuscript, they were also used for format conversion and mechanical consistency checks against the submission template.

All participant data were collected by the author under the approved protocol; AI did not generate, replace, or modify any observation. The research design, methodological choices, interpretation of results, research content, argumentative structure, and principal text of the thesis were written by the author. The author reviewed each AI suggestion and decided whether to adopt it, and verified all citations, numerical values, and statements used in the final text against the original materials. The author accepts responsibility for the complete research project and thesis.

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A Software Environment

All analyses were run in R 4.6.1 on Windows 11. The complete `sessionInfo()` output is reproduced below so that the package versions behind every reported number are on the record.

```
R version 4.6.1 (2026-06-24 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 22631)

Matrix products: default
LAPACK version 3.12.1

locale:
[1] LC_COLLATE=Chinese (Simplified)_China.utf8
[2] LC_CTYPE=Chinese (Simplified)_China.utf8
[3] LC_MONETARY=Chinese (Simplified)_China.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=Chinese (Simplified)_China.utf8

time zone: Europe/London
tzcode source: internal

attached base packages:
[1] stats      graphics  grDevices  utils      datasets  methods   base

other attached packages:
[1] emoa_0.5-3      moocore_0.3.2    rmcrr_0.7.0      patchwork_1.3.2
[5] ggstatsplot_1.1.0 ARTool_0.11.2    effectsize_1.0.3  performance_0.17.1
[9] parameters_0.29.2 emmeans_2.0.4    lmerTest_3.2-1    lme4_2.0-6
[13] Matrix_1.7-5    purrr_1.2.2      tidyr_1.3.2      dplyr_1.2.1
[17] colleRstats_0.1.6 ggplot2_4.0.3

loaded via a namespace (and not attached):
[1] Rdpack_2.6.6      mnormt_2.1.2      polynom_1.4-1
[4] gridExtra_2.3.1   rematch2_2.1.2    sandwich_3.1-3
[7] rlang_1.3.0        magrittr_2.0.5    multcomp_1.4-32
[10] PMCMRplus_1.9.12  ote1_0.2.0        matrixStats_1.5.0
[13] compiler_4.6.1    statsExpressions_2.1.0 BWStest_0.2.3
[16] vctrs_0.7.3       kSamples_1.2-12   quantreg_6.1
[19] stringr_1.6.0     pkgconfig_2.0.3   fastmap_1.2.0
[22] backports_1.5.1   labeling_0.4.3    rmarkdown_2.31
[25] nloptr_2.2.1      MatrixModels_0.5-4 xfun_0.60
[28] cachem_1.1.0      Rmpfr_1.1-2       SuppDists_1.1-9.9
[31] gmp_0.7-5.1       psych_2.6.5       parallel_4.6.1
[34] cluster_2.1.8.2   R6_2.6.1          stringi_1.8.9
[37] RColorBrewer_1.1-3 rpart_4.1.27      car_3.1-5
[40] boot_1.3-32       numDeriv_2016.8-1.1 estimability_2.0.0
[43] knitr_1.51        Rcpp_1.1.2        zoo_1.9-0
[46] DHARMa_0.5.0      base64enc_0.1-6   correlation_0.8.8
[49] nnet_7.3-20       splines_4.6.1     tidyselct_1.2.1
[52] rstudioapi_0.19.0 dichromat_2.0-1   abind_1.4-8
[55] codetools_0.2-20  lattice_0.22-9    tibble_3.3.1
[58] plyr_1.8.9        withr_3.0.3       bayestestR_0.18.1
[61] S7_0.2.2          evaluate_1.0.5    coda_0.19-4.1
[64] foreign_0.8-91    survival_3.8-6    ggpp_0.6.1
[67] RcppParallel_6.2.0 confint_1.0.2     pillar_1.11.1
[70] carData_3.0-6     stats4_4.6.1     checkmate_2.3.4
[73] reformulas_0.4.4  insight_1.5.3    generics_0.1.4
[76] paletteer_1.7.0   rstantools_2.7.0 scales_1.4.0
[79] minqa_1.2.8       xtable_1.8-8     glue_1.8.1
[82] Hmisc_5.2-6       tools_4.6.1      see_0.14.1
[85] ggpmisc_1.0.0     data.table_1.18.4 SparseM_1.84-2
[88] ggsignif_0.6.4    mvtnorm_1.1.2    grid_4.6.1
[91] rbibutils_2.4.1   datawizard_1.3.1 colorspace_2.1-3
[94] nlme_3.1-169      htm1Table_2.5.0  Formula_1.2-6
[97] cli_3.6.6         viridislite_0.4.3 gtable_0.3.6
[100] digest_0.6.39     prismatic_1.1.2  ggrepel_0.9.8
[103] TH.data_1.1-5     htm1widgets_1.6.4 farver_2.1.2
[106] htm1tools_0.5.9   memoise_2.0.1    lifecycle_1.0.5
[109] multcompView_0.1-12 MASS_7.3-65
```

B Post-Session Questionnaire

Table 11 lists the full content of this appendix.

The post-session questionnaire was administered once, after the final round, as a web form rather than inside the headset. It collected the demographics reported in Section 3.2 as well as the items below. Items `design_1` to `design_5` shared the stem *How much do you agree with each statement about the FINAL display design?* and used a five-point scale from strongly disagree to strongly agree. All sixteen participants completed it. Variable names are those used in the released data.

Table 11: Items of the post-session questionnaire, with the variable names used in the released data.

Variable	Item
pcode	Participant code (entered by the experimenter, e.g. P01)
age	Your age (years)
gender	Your gender
yearsdriving	Years of driving experience
drivefreq	How often do you currently drive?
vrxp	Experience with VR headsets
simexp	Experience with steering-wheel simulators
design_1	The final display helped me drive in a more eco-friendly way.
design_2	The final display distracted me from the road.
design_3	Over the rounds, the display changed towards a design that suited me better.
design_4	The final display design suited me.
design_5	I would want a display like this in a real car.
discomfort	During the session I felt dizzy, nauseous, or had eye strain (1 = not at all, 5 = very much)
seatmotion	The motion of the seat felt:
open1	Which display element did you like most, and which annoyed you most?
open2	Is the final design your ideal design? If not, what would your ideal display look like?

C Open-Ended Responses

Table 12 lists the full content of this appendix. Responses originally given in Chinese are presented in the author’s English translation, marked (tr.); the verbatim originals are preserved with the study materials.

The two free-text questions were *Which display element did you like most, and which annoyed you most?* and *Is the final design your ideal design? If not, what would your ideal display look like?* Responses given in English are reproduced verbatim, including spelling and grammar as written, and a dash marks a question left blank.

D Converged Design of Each Participant

Table 13 lists the full content of this appendix.

Each value is the normalised parameter in $[0, 1]$ that the optimiser proposed as that participant’s final design and that was then driven in round 21. The mapping from these values to element sizes and opacity is given in Section 3.4, and an element disappears once its size falls below the legibility threshold.

E Per-Participant Optimisation Progress

Table 14 lists the full content of this appendix.

The hypervolume (HV) columns are taken from the optimiser’s own log, computed in the normalised minimisation space with

Table 12: Verbatim open-ended responses from all sixteen participants.

P	Liked most / annoyed most	Ideal design
P01	leaf instruction	no I want leaf + instruction type
P02	the leaf	yes it is
P03	Like most: Leaf. Annoyed most: The face with the words under it.	Yes.
P04	Smile face	The opacity was slightly too high, though only for the smiley face (tr.)
P05	The leaf with red/green color, and the smile (crying) face with colors (of course the color change is my best preference)	Yes, the number does not change so quickly. So quick changes makes me dizzy and being distracted by the number (i prefer to watch the speed and i think the speed is the most important)
P06	The speed with its colour cue, and the changing face of the eco symbol (tr.)	yes
P07	I think the three-level colour design is good; the eco level can be seen at a glance (tr.)	The ideal design would keep only a single leaf, without other indicators such as speed (tr.)
P08	—	—
P09	The word comments below the Eco-Score display, it can help me know how to improve eco driving.	Nearly the ideal, but I want to keep other car's label
P10	The low transparency, I'd like more information displayed on the front screen, but when they're too much, it distracted me, so low transparency helped reduce that distraction.	Yes
P11	—	YES
P12	I found it quite intuitive that the colour changed on the main speedometer - made it easier to glance at only one thing while driving	Yes it is - I would have appreciated a bit more explanation on what the different components did on my screen because I felt like they changed at different rates or showed different things.
P13	—	—
P14	i like the leaf most, and i do not like the face as it might display negative emotion	i think it is, it's just that i do not care a lot about eco driving but i probably consider using it if environment really needs me to do that
P15	leaf	no, the face emoji will distract me
P16	I liked the leaf in the ECO display, while the text annoyed me the most, as it distracted me from the road.	Yes, it is my ideal design

reference point (1, 1, 1, 1, 1); *HV start* is the value after the fifteen sampling rounds and *HV end* the value after the last optimisation round. The Pareto columns are recomputed independently of the optimiser, a round counting as Pareto-optimal when no other round by the same participant is at least as good on all five objectives and strictly better on one. Rates are per round, over fifteen sampling and five optimisation rounds.

F Script Versions

Table 15 lists the full content of this appendix.

Every session archive carries a snapshot of the scripts that produced it, with modification time, size and hash. The table below is the snapshot taken for the first participant; the released data carry the corresponding per-file hash manifest for every participant, and the verbatim snapshots are preserved in the private archive pending re-identification review. The scripts that changed during data collection, and the sessions on either side of each change, are reported in Section 3.10.

Table 13: The seven design parameters of each participant’s final display.

P	Leaf	Score	Framing	Speed	Accel	Markers	Opacity
P01	0.953	0.666	0.066	0.773	0.382	0.595	0.582
P02	0.646	0.461	0.279	0.716	0.422	0.823	0.845
P03	0.792	0.294	0.343	0.25	0.232	0.806	0.526
P04	0.441	0.263	0.639	1	1	0.398	0
P05	0.554	0.143	0.921	0.528	0.551	0.867	0.713
P06	0.2	0.498	0.564	0.839	0.857	0.396	0.137
P07	0.406	0.248	0.847	0.212	0.622	0.144	0.512
P08	0.607	0.192	0.033	0.115	0.732	0.724	0.729
P09	1	0.216	1	0.152	0.491	0.01	0.639
P10	0.58	0.571	0.548	0.748	0.464	0.205	0.093
P11	1	0.307	0.299	0.081	1	0.071	1
P12	0	0.359	0.311	1	0.275	0.385	0.14
P13	0.798	0.675	0.343	0.092	0.057	0.221	0.105
P14	0.471	0.071	0.125	0.212	0.396	0.683	0.473
P15	0.001	0.161	0.485	0.591	0.421	0.825	0.021
P16	0.552	0.768	0.331	0.386	0.436	0.973	0.331

Table 14: Hypervolume and Pareto counts for each participant.

P	HV start	HV end	Gain	Gain			
P01	10.29	10.29	0.00	0.0	4	0.27	0.00
P02	13.88	23.05	9.17	66.1	9	0.40	0.60
P03	19.42	19.44	0.02	0.1	8	0.47	0.20
P04	21.22	22.41	1.19	5.6	5	0.13	0.60
P05	15.94	15.96	0.01	0.1	7	0.40	0.20
P06	8.24	12.34	4.10	49.7	3	0.00	0.60
P07	13.00	16.20	3.20	24.6	7	0.27	0.60
P08	5.04	5.20	0.16	3.2	9	0.47	0.40
P09	10.48	15.99	5.51	52.5	9	0.33	0.80
P10	15.57	17.80	2.23	14.3	2	0.07	0.20
P11	20.11	20.16	0.04	0.2	2	0.07	0.20
P12	24.95	25.92	0.97	3.9	5	0.27	0.20
P13	19.89	21.38	1.50	7.5	7	0.27	0.60
P14	7.95	10.74	2.79	35.0	8	0.33	0.60
P15	11.83	14.21	2.37	20.1	5	0.07	0.80
P16	7.01	8.01	1.00	14.3	3	0.07	0.40

Table 15: Simulator scripts at the time of the first session, with size and hash.

Script	Modified	Bytes	Hash
StudySetup.cs	2026-08-19 15:06	20028	09341EB50165
EcoFeedbackHUD.cs	2026-08-19 18:43	10554	92EE4A897C3B
EcoScore.cs	2026-08-19 12:47	9817	DA23A2FD08E6
RouteSet.cs	2026-08-19 12:47	3398	7BEDDD468CD9
RoundController.cs	2026-08-19 12:47	10089	C70A623FC4F9
StudyLogger.cs	2026-08-19 12:47	9824	B32A96F35A04
YawMotion.cs	2026-08-19 18:24	8725	31EAEA817920
YawDirectLink.cs	2026-08-19 12:47	13238	A8072504134E
CarController.cs	2026-08-19 12:47	30498	8BB6F6BA145F
BoForUnityManager.cs	2026-08-19 12:46	54139	C099F1D7C588

G Study Materials

This appendix contains the participant-facing study materials approved under UCL Research Ethics Committee project 1165: the task instructions as communicated to participants, followed by the original participant information sheet and consent form, reproduced unaltered on the following pages.

G.1 Task Instructions

There was no separate written task-instruction sheet. After giving consent and completing the demographic questionnaire, each participant received a short oral demonstration of the setup, as described in the participant information sheet. The task-relevant content communicated to every participant was as follows.

- You will drive short city rounds in virtual reality (approximately 22 in the information sheet; the realised protocol comprised 21), rating the display after each round.
- The study takes place at UCL and lasts approximately 40 to 60 minutes.
- The simulator records your driving (speed, acceleration, steering, and vehicle position), the resulting energy use, and your questionnaire answers.
- If you feel unwell, you will be asked to stop, remove the headset, and remain seated for a moment or get some air; the study is aborted on discomfort.
- Breaks are provided every 30 minutes or sooner, following the manufacturer’s usage guidance.
- Please do not mention your name or talk about private matters during the session.

G.2 Participant Information Sheet

The original participant information sheet (four pages) is reproduced on the following pages.

Participant Information Sheet

UCL Research Ethics Committee Approval ID Number: __1165__

YOU WILL BE GIVEN A COPY OF THIS INFORMATION SHEET

Title of Study: Sustainable Personalised Driving

Department: UCL Interaction Centre, University College London

Name and Contact Details of the Researcher(s):

Mark Colley Email: m.colley@ucl.ac.uk

Yike Zhang Email: ucab338@ucl.ac.uk

Name and Contact Details of the Principal Researcher:

Mark Colley Email: m.colley@ucl.ac.uk

1. Invitation Paragraph

Before you decide whether to take part it is important for you to understand why we are doing this research and what participation involves. Please take time to read the following information carefully and discuss it with others if you wish. Ask us if there is anything that is not clear or if you would like more information. Thank you for reading this.

2. What is the project's purpose?

This study is being conducted by the University College London (UCL).

Mobility is a key aspect for inclusion. It is also of major importance for industry, and, unfortunately, a significant contributor to climate change. However, certain aspects have to be personalized. How a person drives strongly affects how much energy the same vehicle uses on the same road, and eco-driving displays are one of the cheapest ways to support this. However, a single fixed display design does not suit every driver.

We want to evaluate whether the design of an eco-driving head-up display can be personalised to the individual driver, online, within a single session. For this, we leverage Virtual Reality. You will drive a short city route repeatedly, and after each drive the display design is adjusted based on the ratings you give.

3. Why have I been chosen?

We are looking for people without dementia and severe illnesses. Participants who are pregnant, elderly, or have pre-existing binocular vision abnormalities, psychiatric disorders, heart conditions, or other serious medical conditions cannot participate. Furthermore, those who use medical devices, including but not limited to pacemakers, hearing aids, or defibrillators, or with a history of seizures, epilepsy, photosensitivity, severe dizziness, or loss of awareness triggered by visual stimuli cannot participate.

The participants are required to speak English well enough to understand the questionnaires and to take part in a semi-structured interview.

4. Do I have to take part?

It is up to you to decide whether or not to take part in this study. If you do decide to take part, you will be given this information sheet to keep and be asked to sign a consent form. You can withdraw at any time without giving a reason and without it affecting any benefits that you are entitled to. If you decide to withdraw you will be asked what you wish to happen to the data you have provided up to that point.

5. What will happen to me if I take part?

If you decide to take part:

1. You will be asked to sign a consent form to say that you agree to take part in the study. You can provide written consent prior to the study or orally on the day of the study if you like. You will be given a copy of this to keep for your records.
2. Once you sign a consent form the researcher will ask you to complete a questionnaire about your demographics (e.g., your age, gender, education, professional status), driving experience (e.g., license duration, driving frequency, kilometres driven last year), and opinions on autonomous driving or other mobility modes. You will have the option to leave a question blank or state 'prefer not answer' if you so choose. This information will be used to compare answers from people who are from different backgrounds.
3. Once you have completed the questionnaire, a researcher will give you a short presentation demonstrating the setup. After this presentation, you will drive in Virtual Reality for approximately 22 short rounds, rating the display after each round, followed by a short closing interview.

- The study will take place at UCL and will last for approximately 40 to 60 minutes
- The simulator will record your driving (speed, acceleration, steering, and the position of your vehicle), the resulting energy use, and your answers to the questionnaires.
- If you feel unwell, we will ask you to stop the study, remove the helmet and remain seated for a moment or accompany you to get some air if you experience any discomfort
- We provide breaks every 30 minutes or shorter, according to the manufacturer's usage descriptions

6. Will I be recorded and how will the recorded media be used?

You will not be audio- or video-recorded at any point. Only the driving data listed above, your questionnaire answers and written notes from the closing interview are recorded. No data will leave UCL premises.

7. What are the possible disadvantages and risks of taking part?

Virtual Reality environments can cause physical discomfort including nausea, eye strain, and disorientation. If this happens, we will abort the study.

8. What are the possible benefits of taking part?

There will be no immediate benefits to participating in the study. However, we hope to use the information we learn from this user study to make improvements to future mobility to make it more accessible and user-friendly.

Participation is unpaid: no payment or voucher is provided for taking part in this study.

9. What if something goes wrong?

If you have a concern about any aspect of this study, you should ask to speak with the researcher, Dr Mark Colley (email: m.colley@ucl.ac.uk) who will do his best to answer your questions. If your concerns are not yet resolved, you can contact the UCL Ethics committee at ethics@ucl.ac.uk.

10. Will my taking part in this project be kept confidential?

All information that is collected from you during the research will be kept strictly confidential and will be stored in accordance with the General Data Protection Regulation (GDPR) 2018. The only people in UCL who will have access to information that identifies you will be people who need to contact you to arrange the research appointments or audit the data collection process.

Your data will be held confidentially, with access restricted to the research team working on the project. The information we collect will be stored on computers at UCL in a form in which you cannot be identified. Only the study research team will have access to the data. The data are retained until they have been analysed and the study has ended. Only anonymized data will be made openly available via GitHub or similar sites.

11. Limits to confidentiality

Please note that assurances on confidentiality will be strictly adhered to unless evidence of wrongdoing or potential harm is uncovered. In such cases UCL may be obliged to contact relevant statutory bodies/agencies.

12. What will happen to the results of the research project?

We will publish the results in reports and papers and talk about the results at conferences, journals, seminars and teaching. Please tell the researchers if you would like a copy of any publications and we would be happy to send them to you when they are published. You will not be identified in any report/publication. With your permission, the data collected during the course of the project might be used for additional or subsequent research. Anonymised data will then be archived for 20 years in accordance with UCL guidelines.

13. Local Data Protection Privacy Notice

The controller for this project will be UCL. The UCL Data Protection Officer provides oversight of UCL activities involving the processing of personal data, and can be contacted at data-protection@ucl.ac.uk

This 'local' privacy notice sets out the information that applies to this particular study. Further information on how UCL uses participant information can be found in our 'general' privacy

notice: <https://www.ucl.ac.uk/legal-services/privacy/ucl-general-privacy-notice-participants-and-researchers-health-and-care-research-studies>

The information that is required to be provided to participants under data protection legislation (GDPR and DPA 2018) is provided across both the 'local' and 'general' privacy notices. The lawful basis that will be used to process your personal data are: "Public task" for personal data and "Research purposes" for special category data.

Your personal data will be processed so long as it is required for the research project. If we are able to anonymise or pseudonymise the personal data you provide we will undertake this, and will endeavour to minimise the processing of personal data wherever possible.

If you are concerned about how your personal data is being processed, or if you would like to contact us about your rights, please contact UCL in the first instance at data-protection@ucl.ac.uk.

Please do not mention your name or talk about private matters.

14. Contact for further information

Dr Mark Colley
UCL Interaction Centre
m.colley@ucl.ac.uk
66-72 Gower St
London WC1E 6EA

Thank you for reading this information sheet and for considering to take part in this research study. *You will be given a copy of this information sheet and a signed consent form to keep.*

G.3 Consent Form

The original consent form (three pages) is reproduced on the following pages.

CONSENT FORM FOR ADULTS IN RESEARCH STUDIES

Please complete this form after you have read the Information Sheet and/or listened to an explanation about the research.

Title of Study: Sustainable Personalised Driving

Department: UCL Interaction Centre

Name and Contact Details of the Researcher(s):

Yike Zhang Email: ucab338@ucl.ac.uk

Name and Contact Details of the Principal Researcher:

Mark Colley Email: m.colley@ucl.ac.uk

Name and Contact Details of the UCL Data Protection Officer:

Alexandra Potts Email: data-protection@ucl.ac.uk

This study has been approved by the UCL Research Ethics Committee: Project ID number: 1165

Thank you for considering taking part in this research. The person organising the research must explain the project to you before you agree to take part. If you have any questions arising from the Information Sheet or explanation already given to you, please ask the researcher before you decide whether to join in. You will be given a copy of this Consent Form to keep and refer to at any time.

I confirm that I understand that by ticking/initialling each box below I am consenting to this element of the study. I understand that it will be assumed that unticked/initialled boxes means that I DO NOT consent to that part of the study. I understand that by not giving consent for any one element that I may be deemed ineligible for the study.

		Tick Box
1.	*I confirm that I have read and understood the Information Sheet for the above study. I have had an opportunity to consider the information and what will be expected of me. I have also had the opportunity to ask questions which have been answered to my satisfaction	
2.	*I consent to participate in the study. I understand that my personal information including, my name, age, gender, email address and/or telephone number, will be used for the purposes explained to me. I understand that according to data protection legislation, 'public task' will be the lawful basis for processing and 'research purposes' will be the lawful basis for processing special category data.	
3.	Use of the information for this project only * I understand that my personal information will remain confidential as far as possible, and that all efforts will be made to ensure I cannot be identified. I understand that my data gathered in this study will be stored pseudo-anonymously and securely in accordance with General Data Protection Regulation (GDPR) and the Data Protection Act 2018., and only the researchers involved in the research project will have access to the data. Derivations from this are highlighted in the next entries. It will not be possible to identify me in any publications.	
4.	*I understand that my information may be subject to review by responsible individuals from the University College London (the study sponsor) for monitoring and audit purposes.	
5.	*I understand that my participation is voluntary and that I am free to withdraw at any time during the user study without giving a reason.	

6.	I understand the potential risks of participating and the support that will be available to me should I become distressed during the course of the research.	Zhang
7.	I understand the direct/indirect benefits for wider society of participating.	
8.	I understand that the data (beyond the anonymized questionnaires) will not be made available to any commercial organisations but is solely the responsibility of the researcher(s) undertaking this study.	
9.	I understand that I will not benefit financially from this study or from any possible outcome it may result in in the future.	
10.	I understand that participation in this study is unpaid, and that no compensation is provided, including if I choose to withdraw.	
11.	I agree that my pseudonymised research data (i.e., the driving data recorded by the simulator and my questionnaire responses) may be used by others for future research. No one will be able to identify you when this data is shared.	
12.	I understand that the information I have submitted will be published as a report. I wish to receive a copy of it.	☐ Yes ☐ No ☐ Yes ☐ No
13.	I hereby confirm that I understand the inclusion criteria as detailed in the Information Sheet and explained to me by the researcher.	
14.	I hereby confirm that: (a) I understand the exclusion criteria as detailed in the Information Sheet (i.e., no dementia and severe illnesses; not pregnant, elderly, or have pre-existing binocular vision abnormalities, psychiatric disorders, heart conditions, or other serious medical conditions cannot participate; not using medical devices, including but not limited to pacemakers, hearing aids, or defibrillators, or with a history of seizures, epilepsy, photosensitivity, severe dizziness, or loss of awareness triggered by visual stimuli) and explained to me by the researcher; and (b) I do not fall under the exclusion criteria.	
15.	I am aware of who I should contact if I wish to lodge a complaint.	
16.	I voluntarily agree to take part in this study.	
17.	Use of information for this project and beyond I would be happy for the data I provide to be archived at UCL. I understand that other authenticated researchers will have access to my pseudonymised data.	
18.	I understand that the researcher will not pass to any third party any of the personal information I provided them with. However, I understand that I should not provide any personal information (e.g., names or addresses) or sensitive personal account (e.g., information about an individual's racial or ethnic origin, political opinions, religious beliefs, trade union membership) of myself or others to limit the possibility to be identified by third parties.	

If you would like your contact details to be retained so that you can be contacted in the future by UCL researchers who would like to invite you to participate in follow up studies to this project, or in future studies of a similar nature, please tick the appropriate box below.

☐	Yes, I would be happy to be contacted in this way	
☐	No, I would not like to be contacted	

Name of participant

Date

Signature

Researcher

Date

Signature